



Drought resistance strategies in minor millets: a review

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Abstract

Main conclusion The review discusses growth and drought-response mechanisms in minor millets under three themes: drought escape, drought avoidance and drought tolerance.

Abstract Drought is one of the most prominent abiotic stresses impacting plant growth, performance, and productivity. In the context of climate change, the prevalence and severity of drought is expected to increase in many agricultural regions worldwide. Millets (coarse grains) are a group of small-seeded grasses cultivated in arid and semi-arid regions throughout the world and are an important source of food and feed for humans and livestock. Although minor millets, i.e., foxtail millet, finger millet, proso millet, barnyard millet, kodo millet and little millet are generally hardier and more drought-resistant than cereals and major millets (sorghum and pearl millet), understanding their responses, processes and strategies in response to drought is more limited. Here, we review drought resistance strategies in minor millets under three themes: drought escape (e.g., short crop cycle, short vegetative period, developmental plasticity and remobilization of assimilates), drought avoidance (e.g., root traits for better water absorption and leaf traits to control water loss), and drought tolerance (e.g., osmotic adjustment, maintenance of photosynthetic ability and antioxidant potential). Data from ‘omics’ studies are summarized to provide an overview of the molecular mechanisms important in drought tolerance. In addition, the final section highlights knowledge gaps and challenges to improving minor millets. This review is intended to enhance major cereals and millet per se in light of climate-related increases in aridity.

Keywords Drought · Climate change · Food security · Minor millets · Omics · Genome editing

Introduction

The rise in atmospheric carbon dioxide (CO₂) concentrations arising from human activity contributes to climate change, which manifests itself in elevated temperatures and

alterations in precipitation patterns, among other effects. Periods of low rainfall, high temperatures, high evaporative demand, and low soil moisture storage capacity can lead to drought. The prevalence and severity of droughts are projected to become more frequent in agricultural regions worldwide. Today, 150 crop species are cultivated extensively, 70% of human calories come from 15 species, and cereal, rice, wheat, and maize contribute over 60% (FAOSTAT 2019). Dependence on a few crop species and a limited genetic basis for selecting the most important cultivars has resulted in food security concerns in the context of rapid climatic change. Although the actual drought-associated yield loss can vary depending on the specific crop and farming practices, global data from 1983 to 2009 indicates that a single drought event can decrease a nation’s gross agricultural production by an average of 0.8% (Kim et al. 2019). Drought stress is estimated to have caused global production losses of \$30 billion over the last decade (Gupta

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et al. 2020), and drought-induced crop losses significantly impact food security and rural livelihoods, particularly in developing countries (Cushman et al. 2022). Moreover, the projected increase in drought frequency and consequential agricultural losses coincide with an increasing human population, predicted to reach 9.7 billion in 2050, necessitating a 70% increase in food production (Vetriventhan et al. 2020). To cope, agricultural practice must evolve by developing improved crops that can maintain yields despite water scarcity and drought.

Millets are grasses known for their small seeds and coarse grains, commonly grown in arid and semi-arid regions worldwide. Millets are part of the PACMAD clade of the Poaceae and are divided into major and minor millets types based on their importance and cultivation scale. Sorghum and pearl millet are major millet crops globally. Minor millets include kodo millet (*Paspalum scrobiculatum* L.), foxtail millet [*Setaria italica* (L.) P. Beauv.], little millet (*Panicum sumatrans* Roth ex Roem. and Schult.), proso millet or broomcorn millet (*Panicum miliaceum* L.), and barnyard millet (*Echinochloa frumentacea* L.) of the subfamily Panicoideae, while, finger millet [*Eleusine coracana* (L.) Gaertn] and teff (*Eragrostis tef*) are in the subfamily Chloridoideae (Vetriventhan et al. 2020). Cereals such as rice, wheat, barley, oats and rye belong to the different subfamilies of the BOP clade. Major cereal crops were domesticated in mesic environments, favouring intensive selection and breeding for higher yields under mesic conditions. This has led to less genetic variability and increased vulnerability to drought stress, making them less suitable for cultivation in regions with scarce water resources (Wilson and VanBuren 2022).

Millets are a crucial food and feed source, ranking sixth in agricultural production worldwide, with Asia and Africa contributing over 95% of total production (Kumar et al. 2018). Minor millets are grown in a smaller area, often as a secondary crop by subsistence farmers in dry and marginal environments. Minor millet grains are superior to that major millets due to their high micronutrient content, particularly calcium and iron, high dietary fiber, essential amino acid content, antioxidant content, gluten-free status, low glycaemic index, and resistance to storage pests. They are recommended as part of a healthful diet for people suffering from gluten intolerance, celiac disease, and diabetes. Additionally, they can be stored for extended periods with little or no loss in quality (Saleh et al. 2013). The UN-World Food Programme has recognized minor millets as future crops for food and nutritional security in hunger-stricken areas. The UN's general assembly has declared 2023 the International Year of Millets, highlighting their potential to improve food security, nutrition, and livelihoods, especially in dryland and marginal areas (Muthamilarasan and Prasad 2021).

Millet species and genotypes exhibit varying degrees of drought-hardiness due to differences in growth responses

and resistance strategies. Major millets, like sorghum and pearl millet, are taller, more ecologically dominant, and less stress-tolerant than shorter, more stress-resilient minor millets (Pardo and van Buren 2021). While, the physiological, biochemical, and molecular processes involved in drought tolerance in major millets are well understood (Badiganavar et al. 2018; Singh et al. 2021; Abreha et al. 2022), much less is known about the mechanisms and processes underlying drought tolerance in minor millets (Vetriventhan et al. 2020; Singh et al. 2021). Here, we review drought resistance strategies adopted by minor millets under three sub-themes: escape, avoidance and tolerance to drought. The escape mechanism permits a plant to quickly complete its life cycle before the onset of drought (e.g., short crop cycle, short vegetative period and early flowering time) and the drought avoidance mechanism allows a plant to maintain high water potential during stress by enhancing water absorption (with better root traits) or reducing water loss (with better leaf traits), while drought tolerance means the plant to maintain turgor and continue metabolism in cells even at low water potential primarily by osmotic adjustment, antioxidant potential and sustained photosynthetic activity. Data from 'omics' studies are summarized to provide an overview of the molecular mechanisms important in drought tolerance. The final section highlights knowledge gaps and challenges in improving minor millets. This review is intended to enhance major cereals and millets per se and in light of increasing aridity.

Growth responses to drought stress

The life cycle of minor millets includes three major growth stages (GS): the vegetative stage (GS1), the panicle development/reproductive stage (GS2), and the grain-filling stage (GS3). These stages can be further divided into ten morphologically detailed secondary growth stages following modification of the Zadoks (Z) scale. GS1 is the initial phase of the crop cycle and consists of sub-stages 0 (seed germination), 1 (seedling growth), 2 (tillering), and 3 (stem elongation). GS2 consists of sub-stages 4 (booting stage), 5 (ear emergence/heading), and 6 (flowering). GS3 is the terminal growth stage and consists of sub-stages 7 (milk/grain development), 8 (dough development), and 9 (physiological maturity). Evaluating drought-associated growth responses and resistance mechanisms in minor millets can be challenging due to genetic variation and environmental influences (Bhatt et al. 2011; Gupta et al. 2020). Nevertheless, contrasting drought tolerances have revealed species/genotype-specific differences in drought resistance among minor millets. Laboratory screening (using osmoticum) and field experiments have been used to study differential growth responses to drought at the three growth stages. Results

show differences in growth among genotypes to varying stress levels depending on the growth stage (Table 1). Water deficits during the reproductive and grain-filling stages commonly occur in rain-fed crops like minor millets, with drought impacting reproductive growth more than vegetative (Talwar et al. 2020), and the early stages of GS1 and GS2 being susceptible to drought.

Vegetative phase (GS1)

Polyethylene glycol (PEG) 6000 or mannitol-based lab screening models were used to examine the impact of water deficit on early seedling growth in minor millets. PEG is a non-ionic water soluble polymer that impedes root water absorption due to its reduced osmotic potential (Shivakrishna et al. 2018). PEG-induced osmotic stress in hydro-culture systems offers advantages like reproducibility, simplicity in scoring shoot and root traits, the ability to screen multiple genotypes in a limited space, and precise control over plant mineral nutrition (Huang et al. 2015). Pot experiments are conducted in controlled indoor environments like greenhouses, polyhouses, and growth chambers, while field experiments are conducted in natural environments under rain-free days or by withholding irrigation (Marchin et al. 2020).

Drought stress inhibits seed germination by disrupting osmotic balance and reducing turgor pressure. Drought stress also leads to impaired seedling growth, reduced root and shoot biomass, and reduced leaf area, with seedling death being common under water deficit and high-temperature conditions (Ndlovu et al. 2021). Although millets require less water for seed germination than other grain crops, drought-induced water stress affects early seedling growth and establishment in a species/genotype-specific manner. Studies show that increased water stress as a function of increased osmotic (PEG or mannitol) concentrations reduces germination, seedling vigor, shoot growth, and fresh and dry weights in finger millet (Mukami et al. 2019; Mundada et al. 2020; Bangari et al. 2023), foxtail millet (Francis et al. 2021), proso millet (Yuan et al. 2022), little millet (Francis et al. 2021), barnyard millet (Singh et al. 2018) and kodo millet (Vikrant and Roobavathi 2021).

During late GS1, plants develop leaves, stems, and roots to absorb water and nutrients and to support photosynthesis, with drought-susceptible cultivars experiencing greater effects on vegetative growth due to water stress. Drought stress increases root growth and the root-to-shoot biomass ratio, so genotypes with more well-developed root systems can mitigate water stress by more effectively accessing water and nutrient resources in deeper soil layers (Abreha et al. 2022). In finger millet, drought stress decreased the leaf area index, the specific leaf area, plant height, and the number of productive tillers (Mwangoe et al. 2022), while mean leaf

temperature and thickness increased in response to water stress (Sanjeev Krishna et al. 2021). Increased leaf temperature under drought stress may be due to decreased leaf water potential and stomatal conductance, resulting in lower evapotranspirative cooling (Mohanabharathi et al. 2019). Relative to thinner leaves, thicker leaves have more mesophyll-containing high-density chlorophyll and are expected to increase the photosynthetic rate at a constant leaf surface area (Craufurd et al. 1999; Balsamo et al. 2006). Productive tillers, which produce mature ear heads during the reproductive phase, are likely to decrease under drought stress due to late-formed ear abortion, affecting photosynthate availability and grain yield (Ludlow and Muchow 1990). These findings underscore the importance of traits at the early stages of GS1 when screening for differential drought tolerance in minor millets (Naik et al. 2020).

Reproductive growth/panicle development phase (GS2)

Previous studies have discovered differences in reproductive growth and development among millets in response to water stress. A study comparing drought-stressed barnyard millet, finger millet, and little millet found differences in overall productivity, the number of productive tillers, the number of ear heads, and ear head length, with barnyard millet and little millet performing best and worst, respectively, and performance differences being possibly due to variable crop cycle duration (Ashok et al. 2018). Similarly, differences in flowering phenology have been discovered between water-stressed millets and millet genotypes (Sanjeev Krishna et al. 2021), with drought and high temperature (40 °C) resulting in delayed flowering (Yogeesh et al. 2016) or increased floret death (Seghatoleslami et al. 2008). A reduction in average ear weight, finger number per ear, finger length, and spikelet fertility has been demonstrated in finger millet in response to water stress (Nielsen and Vigil 2017).

Grain filling stage (GS3)

Drought stress in finger millet reduced grain yields by up to 18% (Nanja Reddy et al. 2020), with stress occurring during the 15-day interval before flowering and 10–15 days after flowering reducing yields 6 and 13.2%, respectively (Nanja Reddy 2020). The positive correlations between tiller number, finger length, and grain yield in finger millet grown under drought conditions (Nanja Reddy et al. 2020; Ramya and Nanja Reddy 2022), while Mwangoe et al. (2022) found a positive relationship between harvest index (HI) and yield. Similarly, Sanjeev Krishna et al. (2021) found that drought stress reduces finger millet grain yield by 36.6%, while Talwar et al. (2020) found drought-associated reductions in HI and grain yield (–45%) in foxtail millet. Grain yields

Table 1 Summary of previously published growth response studies of minor millets under drought stress

Major Growth Stage ^a	Secondary growth stage ^b	Growth Response to drought stress	References
Vegetative Phase (GS1)	0; Germination	Decrease in germination frequency and rate, germination index decreases	Francis et al. (2021), Singh et al. (2018), Vikrant and Roobavathi (2021)
	1; Seedling growth and establishment	Decrease in seedling vigour and plant height Decrease in leaf number, area, and thickness Decrease in plant weight and total biomass Increase in root growth, root/shoot ratio Increase in leaf temperature, specific leaf weight, vein number, and the leaf area/root dry weight ratio	Mukami et al. (2019), Mundada et al. (2020), Naik et al. (2020), Francis et al. (2021), Mwangoe et al. (2022), Ahmad et al. (2019), Ajithkumar and Panneerselvam (2014), Singh et al. (2018), Vikrant and Roobavathi (2021), Senthil et al. (2018), Matsuura et al. (2012), Grašič et al. (2019)
Reproductive Growth Phase (GS2)	2; Tillering: Leaf elongation, Tiller emergence and development	Tillering increases, Productive tillers decrease	Nanja Reddy (2020), Nanja Reddy et al. (2020), Mwangoe et al. (2022), Sanjeev Krishna et al. (2021)
	3; Stem Elongation: stem elongation; panicle initiation	Decrease in shoot growth and biomass Delay in panicle initiation	Talwar et al. (2020), Ahmad et al. (2019), Begum et al. (2013), Ali Karimian et al. (2021)
	4; Booting: 25% of inflorescences emerge; spikelets in the fingers compactly arranged, florets not identifiable	Decrease in panicle number and height Decrease in spikelet fertility	Sanjeev Krishna et al. (2021), Matsuura et al. (2012), Ali Karimian et al. (2021)
	5; Ear Emergence: ears initiate grow and development	Decrease in the number of fingers per ear, finger length, and total and mean ear weight	Ashok et al. (2018), Sanjeev Krishna et al. (2021), Mwangoe et al. (2022)
	6; Flowering: Initiation of anthesis, inflorescences appear, anthers emerge, florets become identifiable	Increase in floret death	Seghatoleslami et al. (2008) Sanjeev Krishna et al. (2021)
	7; Grain/Milk Development; fertilization, grains form, milk endosperm development	Decrease in the grain number per panicle, and grain weight yield	Seghatoleslami et al. (2008) Dai et al. (2012)
Grain filling Phase (GS3)	8; Dough Development: grains progress through the dough stage; dry matter accumulation occurs	Decrease in 1000-seeds weight Decrease in harvest index	Matsuura et al. (2012) Nielsen and Vigil (2017)
	9; Physiological Maturity: Grains reaches full maturity and are harvestable		Lapinmakuni et al. (2018), Nematpour et al. (2019), Grašič et al. (2019), Mohanabharathi et al. (2019), Nanja Reddy et al. (2020), Talwar et al. (2020), Mwangoe et al. (2022), Matsuura and An 2020, Ali Karimian et al. (2021)

^aLife cycle classified into three major growth stages according to the Zadoks (Z) scale^bGrowth stages further classified into ten secondary growth stages according to slightly modified the Zadoks (Z) scale

in *S. italica*, *P. milaceum*, and *P. sumatrans* under water stress decreased by 80, 36, and 20%, respectively, due to fewer grains per panicle and lower total grain dry weight (Matsuura et al. 2012). In proso millet, HI and transpiration efficiency are the main factors influencing grain yield under water deficit (Grašič et al. 2019).

Grain yields in *Panicum* species are susceptible to pre- and post-heading drought (Matsuura and An 2020; Matsuura et al. 2012). Water stress before heading in *foxtail millet* and before and after heading in *proso millet* and *little millet* led to decreased grain weight per plant. In foxtail millet, drought stress at the onset of flowering reduced the total grain weight per plant by 87% compared to well-watered controls (Lapuimakuni et al. 2018). Similarly, grain yields decreased in finger millet and teff in response to pre- and post-heading drought (Maqsood and Ali 2007).

Drought-stress reduces grain yield (decreased seed size, number, and weight) and nutritional quality (Saleh et al. 2013). In a comparative study of little millet, barnyard millet, and finger millet, little millet had the highest reduction in 1000-grain weight under drought stress, followed by barnyard millet and finger millet (Ashok et al. 2018). In proso millet, drought stress decreases seed size, reducing HI for both ear and seeding per cent (Seghatoleslami et al. 2008).

Drought resistance strategies in minor millets

Millets have evolved strategies to resist drought stress, including drought escape, avoidance, and tolerance. Responses to drought stress vary from the molecular to whole plant levels and are determined by the plant species, development stage, and drought intensity (Levitt 1980; Kooyers 2015). Figure 1 summarizes drought escape, avoidance, and tolerance mechanisms in minor millets.

Drought escape mechanisms

Drought escape (DE) strategies involve either completion of the life cycle before the onset of severe drought or changes in the rate of growth and reproduction in response to water stress. Millets have various DE mechanisms, including short crop cycles, developmental plasticity, and remobilization of assimilates to seeds. With the exception of short crop cycles, these mechanisms have been well-studied in major millets but are relatively uncharacterized in minor millets (Badiganavar et al. 2018; Vetriventhan et al. 2020).

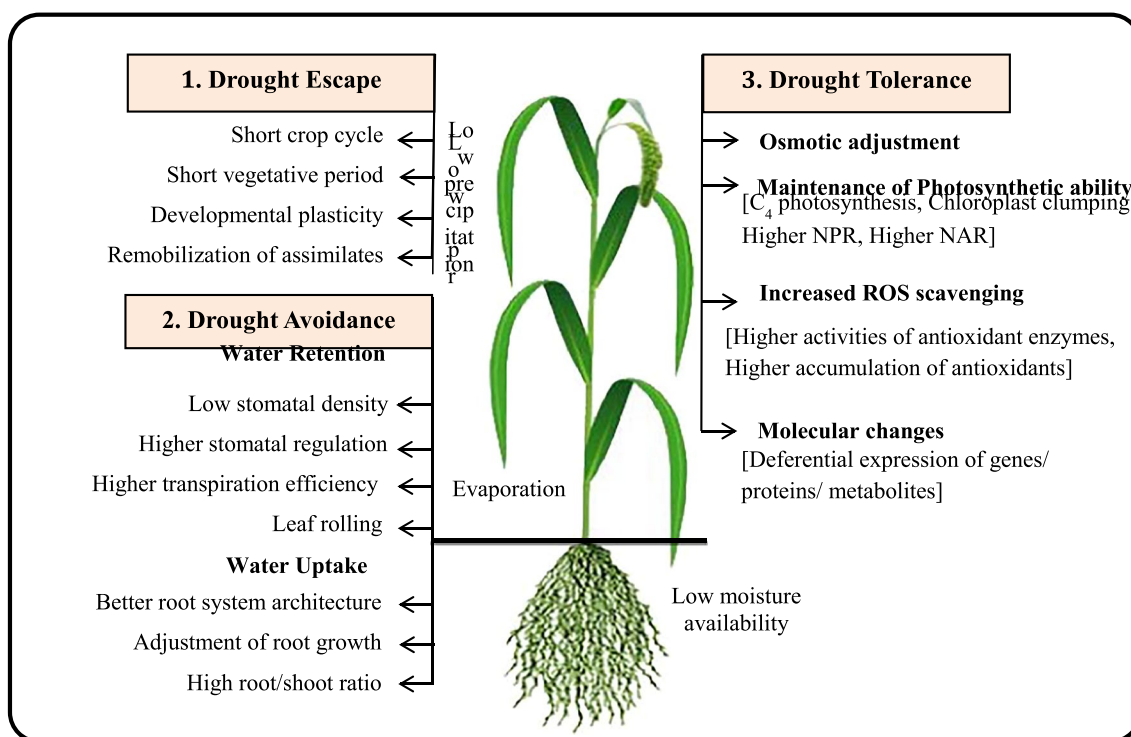


Fig. 1 Drought resistance strategies in minor millets

Short crop cycles

Millet crop duration cycles range from 70 to 115 days, whereas cereals such as rice and wheat have crop duration cycles exceeding 120 days. Sorghum, finger millet and kodo millet have medium crop duration cycles ranging from 105 to 115 days, while pearl millet, foxtail millet, little millet, proso millet, and barnyard millet have shorter cycles ranging from 70 to 90 days (Kumar et al. 2018). More rapid progression through the life cycle allows plants to escape drought stress by completing growth, development, and reproduction before the onset of severe drought. A short vegetative period (SVP), early flowering time (EFT), and early maturity (EM) facilitate rapid growth and completion of the life cycle, increasing survival in drought-prone environments and improving resilience to changing climates (Liu et al. 2020).

The transition from vegetative to reproductive growth is crucial in plant development. It includes tillering, floral initiation and panicle development, flowering, and grain filling. In millets, the timing of floral initiation, also known as heading or ear emergence, is affected by various environmental factors, including temperature, photoperiod, and water availability, and the genetic basis of flowering initiation is well known (Andrés and Coupland 2012). The genetic regulation of accelerated flowering under salt stress reported in foxtail millet, suggests a mechanism for flowering time plasticity in response to water availability (Liu et al. 2020). Phenotypic plasticity in flowering time could be adaptive in hastening the completion of the reproductive cycle under stress while allowing for more complete reproductive growth and development under adequate water availability.

Due to their shorter vegetative period and earlier flowering time, dry land farmers prefer short-duration high-tillering varieties for better yield and forage value. However, the relationship between cycle duration and yield is complex (Matsuura and An 2020; Nanja Reddy et al. 2019); rapid seed germination, early leaf development, and synchronised basal tillering lead to better canopy development, less transpirative water loss, and increased yields in short cycle millets under drought conditions. In addition, short-duration millets generally have seeds that germinate more rapidly, have more profound, more well-developed root systems, and produce leaves earlier (Nanja Reddy 2020). Among minor millets, Pavithra et al. (2020) found that foxtail millet (30.5 g/plant), a short-duration crop, had the highest grain yield, followed by medium-duration finger millet (30.0 g/plant) and kodo millet (26.17 g/plant). Although pearl millet, finger millet, and proso millet had higher total biomass production, grain yields and HI ratios were lower compared to long-duration crops (e.g., wheat, rice, maize, and sorghum) (Kumar et al. 2018).

Developmental plasticity

Millets escape drought by exhibiting plasticity in the growth and development of vegetative and reproductive organs. Growth under water stress results in plants with deeper roots, more erect stems, increased nodal tillering, and increased vegetative biomass. In millets, drought escape strategies also include mid-day leaf folding, a more ascending leaf orientation, reduced chlorophyll biosynthesis, thicker stems, and more rapid recovery following leaf wilting (Sanjeev Krishna and Nanja Reddy 2021; Nanja Reddy 2022). The late-flowering genotypes of pearl millet capture more light, grow taller, and make more tillers compared to earlier flowering varieties (Diack et al. 2020). Leaf orientation influences light capture and photosynthetic efficiency, with increasing leaf erectness decreasing incident light capture but improving light use efficiency (Niinemets 2010). The relationship between incident light intensity and photosynthetic rate is non-linear, with erect upper leaves capturing less light but lower leaves receiving more light (Long et al. 2006).

Remobilization of assimilates

Drought stress influences the size of source and sink tissues and hinders phloem loading, affecting assimilate translocation and dry matter partitioning. Depending on species, growth stage, and stress duration and severity, plants remobilize assimilates stored in leaves, roots, and grains in supporting root growth, thus enhancing water uptake. Upon stress alleviation, millets respond by remobilizing pre-stored carbohydrates from the stem to the grain, stabilizing grain yield and contributing to grain filling (Leport et al. 2006).

Drought avoidance mechanisms

Drought avoidance (DA) is the ability to maintain a positive water balance despite soil moisture decline. Root and leaf modifications avoid drought stress and exhibit considerable variation among minor millet species (Ajithkumar and Panneerselvam 2014). Root architecture, root growth, and root-to-shoot biomass influence water absorption, while leaf traits such as shape, surface area, and stomatal conductance affect water loss. The co-occurrence of root and shoot adaptations and root–shoot functional coordination improves water absorption and decreases water loss under drought conditions.

Root growth

Root growth in plants is influenced by genotype, water availability, soil type, and developmental age, among other factors. Plants extract water from shallower soil zones (e.g., 15–30 cm depth) under mild water stress and deeper zones

(e.g., 30–45 and 45–60 cm depths) when water availability is more limited. Thus, genotypes with more well-developed root systems can mitigate drought effects by accessing water and nutrients from deeper soil layers. Compared to cereals, minor millets such as little millet (Ajithkumar and Panneerselvam 2014; Francis et al. 2021), kodo millet (Francis et al. 2021), barnyard millet (Singh et al. 2018), finger millet (Mundada et al. 2020; Naik et al. 2020; Francis et al. 2021; Nanja Reddy 2020) and foxtail millet (Francis et al. 2021) increase root growth and develop deeper root systems under water stress, with the most drought tolerant genotypes growing 26% deeper roots 15–30 days after sowing (Comas et al. 2013). Furthermore, increases in root length correlate strongly with increased root biomass, which is critical to the transition from vegetative to reproductive growth during development (Sanjeev Krishna et al. 2021).

Root-to-shoot (R/S) ratio

The root/shoot ratio is an important determinant of millet seedling survival (Sanjeev Krishna and Nanja Reddy 2021), with plants disproportionately allocating resources to root growth and disproportionately reducing shoot growth under water stress (Singh et al. 2018; Mundada et al. 2020). In finger millet, the root/shoot ratio on a length basis is maximized 15–30 days after sowing under moisture deficit, compensating for the root volume. In contrast, after stress alleviation, the root/shoot ratio increases on a dry mass basis (Sanjeev Krishna and Nanja Reddy 2021).

Leaf traits for decreasing water loss

Leaf anatomy, morphology, and phenology significantly influence drought avoidance in minor millets. Anatomical features such as leaf thickness and epidermal trichomes affect water use efficiency, reduce transpiration, and increase water retention. Cuticular wax reduces non-stomatal water loss in minor millets and enhances photoprotection, aiding drought avoidance (Pan et al. 2016). Millets produce thicker leaves with more epidermal trichomes and thicker epidermal cuticles, limiting water loss. The composition and amount of epicuticular wax varies depending on species, organ, and developmental state (Nanja Reddy 2020).

Total leaf surface area impacts light interception, carbon assimilation, and evapotranspirative loss. Due to tradeoffs among light absorption, gas exchange, water balance, and productivity, it is a crucial parameter at the nexus of drought stress and yield. A larger leaf area allows for greater light capture, increased photosynthetic rate, and greater productivity while increasing evapotranspirative loss and requiring greater water uptake. Although smaller total leaf surface areas are typically smaller, millets exhibit leaf area plasticity depending on soil moisture availability, leaf turgor

pressure, canopy temperature, and photoassimilate availability (Esmaeil Zade-Moridani et al. 2015). In finger millet, the impact of drought on leaf area is genotype dependent with stress from 18 days before to 10 days after flowering reducing the leaf area by 8.8% (Nanja Reddy 2022) and 16.5% (Sanjeev Krishna et al. 2021). Reduction in leaf area may be due to decreased leaf turgor or a reduction in cell size and number during leaf growth and development (Sanjeev Krishna et al. 2021; Nanja Reddy 2022).

The three-dimensional architecture of a leaf is influenced by physiological processes like photosynthesis, transpiration rate, and water use efficiency. The optimal size and shape of leaves help in efficient light-harvesting, water collection, and faster growth rates and increased yield. Drought can negatively affect leaf shape and size, reducing assimilation and stomatal conductance and leading to decreased yield (Parvathi et al. 2013). To compensate for this, several morphological adaptations, such as leaf rolling, have evolved due to a change in the leaf water potential (Kadioglu et al. 2012). The leaf rolling is controlled by the regulation of by *ADL1*, *RFS* genes of leaf polarity, *SRL1*, *ROC5*, *OsRRK1*, *SLL2*, *CLD1*, *OsZHD1/2*, and *NRL1* genes of bulliform and *SLL1* and *SRL2* genes of sclerenchyma cell formation and *CFL1*, and *CLD1* genes of cuticle development in rice (Xu et al. 2018). Leaf rolling is controlled by the climate and hence follows a diurnal cycle. Leaves begin to unroll at dawn, continue to roll until midday, and then resume unrolling in the afternoon due to a reduction in solar radiation and an air vapour pressure deficit (Ben-Amar et al. 2020). Leaf rolling was observed in foxtail millet (where leaves have thin midrib with higher leaf width) and finger millet (where leaves have thicker midrib and narrow leaf width) in response to water stress that occurs when leaf water potential is -5.0 bars or less (Suma 2014). Wheat crop leaves display rolled-type dynamics under water stress, varying from tightly rolled to loosely roll depending on stress and environmental conditions. This architecture enhances input use efficiency, and photosynthetic machinery efficiency, assimilates translocation, and biomass, and compensates for drought stress yield losses (Ali et al. 2022).

Transpiration efficiency and stomatal conductance

Transpiration efficiency is the ratio of carbon fixed to transpirative water loss during photosynthesis. Transpiration efficiency is important to plant survival as it relates water-use and photosynthetic efficiencies to growth and development. Millets can better tolerate drought stress due to higher transpiration efficiencies that positively correlate with low stomatal densities, high stomatal conductance, and enhanced stomatal regulation, resulting in reduced transpirative loss (Esmaeil Zade-Moridani et al. 2015; Maai et al. 2020; Pavithra et al. 2020; Nanja Reddy et al. 2020). Lower leaf

areas and higher thicknesses also contribute to reduced transpirative losses, higher transpiration efficiencies, and higher productivity in millets grown in dry environments (Balsamo et al. 2006; Vadez et al. 2021).

Leaf phenology

Leaf phenology, including lifespan and senescence, impacts tolerance to drought stress. Leaf senescence is a unique process characterized by chlorophyll and carotenoid degradation, the production of senescence-promoting phytohormones, and cellular, organellar, and macromolecular loss in mesophyll (Schippers et al. 2015). Leaf senescence is triggered in response to various environmental factors, including drought, with minor millets inducing senescence of lower leaves and moderating leaf production under drought stress (Guo and Gan 2012). The stay-green (STG) trait, or green leaf area at maturity (GLAM), is an integrated drought-adoption trait in millets. This trait is associated with higher chlorophyll content, lower green leaf area loss rates, decreased tillering, and reduced size of upper leaves (George-Jaeggli et al. 2017). Functional STG genotypes maintain higher photosynthetic rates and transpiration efficiencies under drought stress (Vadez et al. 2011). Although well-studied in major millets like sorghum (Borrell et al. 2000) and pearl millet (Yadav et al. 2011), information about STG is relatively limited in minor millets (Dai et al. 2012).

Leaf temperature

Leaf temperature (LT) is a recognised indicator of plant water status (Ramya and Nanja Reddy 2022). Drought stress increases leaf temperatures (Jones and Leinonen 2003) during increased water absorption, high transpirative loss, high stomatal conductance, and heat load reflectance through epicuticular wax lower leaf temperatures. In millets, the measurement of individual leaf temperatures using a simple hand-held infrared thermometer has been used most effectively to rapidly screen genotypes for moisture stress tolerance (Hirayama et al. 2006). Ramya and Nanja Reddy (2018) and Sanjeev Krishna et al. (2021) screened 181 germplasm lines in examining the effects of LT on grain yield in finger millet, reporting a strong negative correlation ($r = -0.694$) between temperature and yield (Sanjeev Krishna et al. 2021; Suma 2014).

Drought tolerance mechanisms

Soil water availability, ranging from highest (field capacity or FC) to lowest (permanent wilting point or PWP), is the amount of water the soil can hold. Water availability decreases linearly as the soil dries, limiting root absorption and water availability to leaves. Relative water content

(RWC) and leaf water potential (LWP) are physiologically relevant integrators of soil water availability. Relative water fluxes influence RWC and LWP within the soil–plant–atmosphere continuum (SPAC) (Jones 2007). Reduced water availability may lead to cellular dehydration, loss of turgor, and cell membrane disintegration with associated decreases in metabolic activity and photosynthetic rate (Mukarram et al. 2021). Drought tolerance is the ability to withstand cellular dehydration and establish new homeostasis over time through physiological, biochemical, and molecular mechanisms. Tolerance involves the perception of a stress signal, multiple transduction pathways, and expression of stress-responsive genes (Lata et al. 2011; Mukarram et al. 2021). Physiological mechanisms contributing to drought tolerance in minor millets are listed in Table 2. Understanding drought tolerance in minor millet is progressing, but knowledge of the molecular mechanisms underlying tolerance is still lacking (Mukami et al. 2019).

Physiological responses

Relative water content Among the different responses of plants to water stress, maintaining higher relative water content (RWC) is one of the important drought tolerance mechanisms. RWC is the ratio of the actual water content to its maximum water-holding capacity (Barrs and Weatherley 1962). Several studies have shown a relationship between RWC and water stress in finger millet (Bartwal et al. 2016; Mohanabharathi et al. 2019; Mukami et al. 2019; Mundada et al. 2020; Naik et al. 2020; Mwangoe et al. 2022), foxtail millet (Xu et al. 2019) and barnyard millet (Singh et al. 2018). Research shows that drought-stressed minor millets have lower RWC than well-watered plants. The experiments on RWC focused on the vegetative stage, and values vary depending on genotype. Drought-tolerant genotypes have higher RWC values (78–79%) compared drought-sensitive genotypes of finger millet (Aparna et al. 2017) and foxtail millet (Dipti et al. 2018). A reduction in RWC (2.58%) due to DS decreased the stomatal conductance (21.7%), transpiration rate (25.3%), and photosynthetic rate (16.8%) in finger millet (Nanja Reddy et al. 2019, 2020). Lower RWC accelerates leaf senescence and shortens the grain filling period, resulting in reduced grain yields. Therefore, RWC could be an appropriate trait for drought avoidance in millets (Nanja Reddy et al. 2019, 2020).

Leaf water potential Leaf water potential (LWP) is a crucial physiological trait for drought tolerance (Nanja Reddy et al. 2020). LWP doesn't differ between tolerant and susceptible genotypes under well-water conditions. However, under water stress, tolerant finger millet genotypes had smaller LWP reductions (6.9%) compared to the susceptible genotypes (14.4%) at an LWP of -5 bars (Suma 2014).

Table 2 Physiological responses contributing to drought tolerance in minor millets

Physiological response/traits	Tolerance mechanism	Reference
Regulation of relative water content (RWC); Relatively high RWC in tolerant genotypes	Maintenance of hydration and overall water status	Bartwal et al. (2016); Satish et al. (2018), Mukami et al. (2019), Naik et al. (2020), Mohanabharathi et al. (2019), Mwango et al. (2022), Xu et al. (2019), Singh et al. (2018)
Control of leaf water potential (LWP); Relatively high LWP in tolerant genotypes	Maintenance of leaf hydration as a measure of overall water status and dehydration avoidance	Matsuura et al. (2012), Suma (2014), Matsuura and An (2019)
Regulation of Osmolyte Concentrations; Accumulation of proline	Accumulate of inorganic and organic osmolytes indicate effective osmotic adjustment Osmotic adjustment assists in maintaining water absorption and cell turgor pressure	Bhatt et al. (2011), Arun et al. (2018), Mohanabharathi et al. (2019), Mundada et al. (2020), Xu et al. (2019), Ajithkumar and Paneerselvam (2014) Mundada et al. (2020), Ajithkumar and Paneerselvam (2014)
Regulation of Osmolyte Concentrations; Accumulation of glycine betaine		Mundada et al. (2020), Naik et al. (2020), Xu et al. (2019)
Regulation of Osmolyte Concentrations; Accumulation of water-soluble sugars; sucrose, fructose, glucose, galactose, raffinose, stachyose		
Regulation of guard cell turgor/stomatal aperture; Relatively high stomatal conductance in tolerant genotypes	Maximizing CO ₂ uptake and minimizing water loss while maintaining photosynthetic efficiency	Esmail Zade-Moridani et al. (2015), Nanja Reddy et al. (2020), Pavithra et al. (2020)
Higher transpiration efficiency; Relatively high efficiencies in tolerant genotypes	Balancing transpiration with photosynthetic efficiency through low stomatal density, rapid stomatal regulation, and efficient water-use mechanisms	Khan et al. (2014), Nanja Reddy et al. (2020), Talwar et al. (2020), Pavithra et al. (2020)
Balance of photosynthetic pigment composition Decrease in chlorophyll a, b, and total chlorophyll Relatively high pigment concentrations in tolerant genotypes	Decreased chlorophyll concentrations as a measure of photo-oxidative stress Higher chlorophyll concentrations correlate with improved photosynthetic efficiencies	Arun et al. (2018); Bartwal et al. (2016), Mukami et al. (2019), Naik et al. (2020), Mwango et al. (2022), Ahmad et al. (2019), Ajithkumar and Paneerselvam (2014), Singh et al. (2018), Dai et al. (2012)
Higher chlorophyll a/b ratio	Disproportionate reductions in Chl <i>b</i> Chl <i>a/b</i> ratio correlates with tolerance	Ajithkumar and Paneerselvam (2014), Singh et al. (2018)
Higher carotenoid concentrations	Disproportionately high carotenoid concentrations as a measure of ROS scavenging efficiency	Vuletić et al. (2019)
Increases in chlorophyll stability; Tolerant genotypes have higher chlorophyll stability indices (CSI)	CSI as a measure of photosynthetic capacity and membrane integrity	Bhatt et al. (2011), Mohanabharathi et al. (2019), Singh et al. (2018)
Maintenance of PSII efficiency; Relatively high PSII efficiencies in tolerant genotypes	Variable to maximum chlorophyll fluorescence ratio (Fv/Fm) as a marker for PSII photoinhibition	Lu et al. (2015), Panda et al. (2021)
The number and distribution of chloroplasts in mesophyll (M) cell and physiological interactions of M and bundle sheath (BS) cells in C ₄ photosynthesis	Centrifugal distribution promotes metabolite exchange between M cells and BS chloroplasts Centripetal distribution minimizes CO ₂ leakage from BS cells to M cells of Chloroplast clumping protects chloroplasts from photo-oxidation damage	Dengler and Nelson (1999), Yamada et al. (2009), Maai et al. (2020)
Three biochemical subtypes of C ₄ photosynthesis (NADP-ME, NAD-ME, and PEPCK)	Increased tolerance and water use efficiencies in NAD-ME crops	Ghannoum et al. (2002)
Plasticity in net assimilation rates (NAR)	Variable NAR as a measure of photosynthetic efficiency in C ₄ subtypes	Matsuura et al. (2012), Senthil et al. (2018)

Table 2 (continued)

Physiological response/traits	Tolerance mechanism	Reference
Variable photosynthetic rate Relatively high rates in tolerant genotypes	Differential photosynthetic ability as a function of C4 subtype of photosynthesis and drought stress	Matsuura et al. (2012), Xu et al. (2006), Dai et al. (2012), Lu et al. (2015), Senthil et al. (2018), Mohanabharathi et al. (2019), Nanja Reddy et al. (2020), Pavithra et al. (2020), Mwangoe et al. (2022), Yuan et al. (2022), Lu et al. (2015)
Scavenging of H ₂ O ₂ and other ROS; Relatively low ROS concentrations in tolerant genotypes	Reduction in oxidative damage to membranes, organelles, and macromolecules	Bhatt et al. (2011), Bartwal and Arora (2016), Satish et al. (2018), Lata et al. (2011), Nematopour et al. (2019), Dai et al. (2012)
Regulation of MDA concentrations Relatively low MDA concentrations in tolerant genotypes	MDA concentration as a measure of ROS presence and oxidative damage	Bhatt et al. (2011), Bartwal and Arora (2016), Mukami et al. (2019), Mundada et al. (2020), Lata et al. (2011), Xu et al. (2019), Nematopour et al. (2019), Dai et al. (2012)
Control of electrolyte leakage; Relatively low electrolyte leakage in tolerant genotypes	Electrolyte leakage as a measure of membrane oxidative damage	Bhatt et al. (2011), Sudan et al. (2015), Satish et al. (2018), Singh et al. (2018), Liu et al. (2020)
Regulation of antioxidative enzyme expression and activity Relative high APX:SOD and APX:MDR ratios in tolerant genotypes	Increased expression of antioxidative enzymes in moderating oxidative damage	Bhatt et al. (2011), Bartwal and Arora (2016), Panda et al. (2021), Lata et al. (2011), Ahmad et al. (2019), Nematopour et al. (2019), Dai et al. (2012)
Regulation of the concentrations of antioxidative metabolites	Higher accumulation of antioxidants as a measure of ROS scavenging efficiency	Ahmad et al. (2019)

Water movement by aquaporins (AQP) Water movement from roots to leaves occurs through three main routes: apoplastic (cell walls), symplastic, and transcellular. The cell-to-cell pathway, which can occur via plasmodesmata or membranes, can be driven by osmotic and hydrostatic forces. Aquaporins (AQP) may be involved in cell-to-cell water transfer, while a hydrostatic pressure gradient in inter-cellular space drives apoplastic water transport. A stress tolerance study by Singh et al. (2019) on foxtail millet, identified several intrinsic membrane proteins expressed in response to water stress. These included 12 intrinsic plasma membrane proteins (PIPs), 11 tonoplast intrinsic proteins (TIPs), 13 NOD26-like intrinsic proteins (NIPs), and three small essential intrinsic proteins (SIPs) (Singh et al. 2019). The expression and functional aspects of AQPs in other minor millets are yet to be determined.

Osmotic adjustment During drought, plants initiate physiological processes such as osmotic adjustment (OA), which helps to maintain water absorption and cell turgor pressure. During OA, plants accumulate large amounts of organic and inorganic osmolytes. This active reduction of osmotic potential is observed in various species, including minor millets. Osmolytes protect proteins, enzymes, nucleic acids, organelles, and membranes from oxidative damage. Inorganic osmolytes, such as potassium, sodium, calcium, chloride, and nitrate ions, have been found to play essential roles in ion transport under stress conditions. Organic osmolytes, like amino acids (e.g., proline) and their analogues (e.g., glycine betaine), water-soluble sugars (WSCs), and sugar alcohols (sorbitol, mannitol) also accumulate under drought stress. However, their importance may vary between genotypes and developmental stages (Mukarram et al. 2021).

Minor millets can successfully adjust osmotic potential in response to drought stress by accumulating proline (Lata et al. 2011; Ajithkumar and Panneerselvam 2014; Bhatt et al. 2011; Mundada et al. 2020; Naik et al. 2020), glycine betaine (Mundada et al. 2020; Ajithkumar and Panneerselvam 2014; Arun et al. 2022) and WSC (Naik et al. 2020; Arun et al. 2022). Proline accumulation is a universal symptom of foliar dehydration linked to stress sensitivity, with higher accumulations in susceptible than tolerant varieties. Many studies on minor millets report relatively high proline accumulation in drought-tolerant genotypes. However, the particular role of proline in drought tolerance remains controversial (Dien et al. 2019). The proline accumulation was found to be an unreliable predictor for drought tolerance in finger millet genotypes (Naik et al. 2020). Transgenic studies show that proline increases are too low to account for overall osmotic adjustment despite potentially high levels in specific cell types or compartments (Zhu 2001).

Drought stress increased WSC concentrations in finger millet during the vegetative (GS1), panicle development/

reproductive (GS2), and grain-filling stages (GS3) (Naik et al. 2020). The sucrose accumulation was higher than that of other WSCs, possibly due to increased import (Hu et al. 2020) or the downregulation of cleaving enzymes (Yang et al. 2019). Sucrose accumulation was higher in drought-tolerant than drought-sensitive genotypes and lower in seedlings than in the other developmental stages (Naik et al. 2020). At lower concentrations, sucrose may function as a substrate for carbohydrate metabolism or signal for stress modifications, while higher concentrations may act as an osmoprotectant (Yang et al. 2019). Raffinose family oligosaccharides (RFOs) are known to perform diverse functions in plants, including biotic and abiotic stress tolerance. The RFOs accumulated in four-finger millet genotypes under drought stress (Naik et al. 2020).

Photosynthetic ability

Photosynthetic activity can be assessed by analyzing chlorophyll and carotenoid concentrations, with decreased chlorophyll indicating oxidative stress, possibly due to photo-oxidation and degradation (Mukami et al. 2019). Plants with higher chlorophyll concentrations under drought stress use light energy more effectively, enhancing drought resistance. Photosynthetic pigment concentration is affected less by stress in drought tolerant genotypes, making the evaluation of chlorophyll concentrations effective in screening of genotypes for differential drought tolerance in finger millet (Bhatt et al. 2011; Arun et al. 2022; Bartwal et al. 2016; Mukami et al. 2019; Mundada et al. 2020; Naik et al. 2020; Mwangoe et al. 2022), foxtail millet (Dai et al. 2012), barnyard millet (Singh et al. 2018), finger millet (Bhatt et al. 2011; Mukami et al. 2019), proso millet (Ahmad et al. 2019) and little millet (Ajithkumar and Panneerselvam 2014). Chlorophyll b (Ch *b*) reductions in millets were disproportionately larger than those of chlorophyll *a* (Chl *a*) under drought stress. The resulting increase in the Chl *a/b* ratio was correlated with improved tolerance in drought-resistant varieties (Singh et al. 2018). Carotenoids, concentrations of which decreased less than those of chlorophyll under drought stress, play a vital role in the ROS scavenging system (Vuletić et al. 2019).

Chlorophyll stability index The chlorophyll stability index (CSI) is a crucial parameter indicating a plant's photosynthetic capacity and membrane integrity (Murty and Majumder 1962). Drought-tolerant genotypes possess lower chlorophyllase activity and higher CSI values than drought-sensitive finger millet (Bhatt et al. 2011; Suma 2014; Aparna et al. 2017; Mohanabharathi et al. 2019; Naik et al. 2020) and barnyard millet (Singh et al. 2018). Higher CSI values suggest stress affects chlorophyll concentrations and photosynthetic activity less in drought tolerant genotypes (Bhatt et al. 2011).

Chlorophyll fluorescence Drought stress negatively impacts the photosynthetic apparatus, reducing the activity of Photosystem II (PSII). The variable to maximal chlorophyll fluorescence (Fv/Fm) ratio is considered a reliable marker for PSII photoinhibition (Lu et al. 2015). A comparative study of a foxtail millet cultivar (Datong29) and hybrid cultivars (Zhangzagu 3, Zhangzagu 5) showed that drought stress resulted in the highest quantum output of PSII photochemistry and the highest excitation capture efficiency of PSII reaction centers. The leaf photochemical activity was assessed using chlorophyll fluorescence in four indigenous finger millet cultivars (Panda et al. 2021). The results showed that drought treatment reduced PS II activity, maximal fluorescence, photochemical efficiency, quantum yield, electron transport rate, and photochemical quenching while increasing minimum fluorescence and non-photosynthetic quenching. The study suggests that chlorophyll content, CSI, and quantum yield of PSII could be effectively used for screening drought-tolerant cultivars of minor millets.

Photosynthetic types C₄ is the primary photosynthetic pathway in minor millets, with higher photosynthetic capacity, water use efficiency, and tolerance in dry environments than C₃ plants. C₄ plants are classified into three biochemical subtypes based on the enzyme that catalyzes decarboxylation: nicotinamide adenine dinucleotide phosphate-dependent malic enzyme (NADP-ME, i.e., sorghum, foxtail millet), nicotinamide adenine dinucleotide-dependent malic enzyme (NAD-ME, i.e., finger millet, little millet, kodo millet, proso millet) and phosphoenolpyruvate carboxykinase (PEPCK, i.e., *Panicum maximum*) (Pardo and VanBuren 2021).

Anatomical, biochemical, and physiological characteristics are associated with C₄ subtypes. C₄ subtypes in millets have parallel venation with major and small veins transporting photosynthates longitudinally and laterally, with small longitudinal veins collecting and transverse veins linking longitudinal veins (Altus and Canny 1982). The variation in the number of leaves and diameter of the prominent and small veins was investigated in minor millets (Parvathi et al. 2020). Foxtail millet had the most major veins, followed by proso millet, finger millet, little millet, barnyard millet, and kodo millet. The total number of small veins was found to be highest in foxtail millet, followed by proso millet, little millet, barnyard millet, finger millet and kodo millet. Barnyard millet had the largest primary vein diameter, followed by finger millet, proso millet, little millet, kodo millet and foxtail millet. Finger millet was found to have the highest small vein diameter, followed by barnyard millet, proso millet, little millet, kodo millet and foxtail millet. According to this study, minor millets have a denser vein system than cereals, which may aid in more efficient water transport and carbohydrate translocation (Parvathi et al. 2020). Among

finger millet genotypes, mean vein frequency (56.8 per leaf width of 1.0–1.4 cm) significantly correlated with photosynthetic rate and grain yield (Nanja Reddy 2020). Higher vein frequency results in more efficient translocation of photo-assimilates to the grain (Vivitha and Vijayalakshmi 2015); therefore, screening genotypes for vein frequency may be useful to programs aimed at increasing the harvest index and grain yield of millets. Variation in the diameter of vascular bundles and bundle sheath cells has also been studied in minor millets. The highest vascular bundle and bundle sheath diameter were found in finger millet and little millet, respectively (Pavithra et al. 2020).

Chloroplast clumping Mesophyll (M) and bundle sheath (BS) cells have numerous chloroplasts, with M cells having chloroplasts randomly distributed along their cell walls. BS chloroplasts are typically located in either the centripetal (close to the vascular tissue, as in finger millet) or centrifugal (close to the MS cells, as in maize) position (Dengler and Nelson 1999). The centrifugal arrangement is advantageous for metabolite exchange between M cells and BS chloroplasts. The centripetal position maximizes CO₂ diffusion between BS and M cells and minimizes CO₂ leakage from BS cells to M cells. These unique arrangements are believed to be mediated by the cytoskeletal network and vascular pressure (Kobayashi et al. 2009). M chloroplasts aggregate to the BS side in response to drought and high-intensity light in finger millet (Maai et al. 2020). The movement of chloroplasts within M is known as chloroplast clumping or chloroplast dispersal, is thought to protect chloroplasts from photooxidative damage during drought, and may support photosynthetic activity under stressful conditions (Maai et al. 2020).

The movement of M chloroplasts towards the BS side in C₄-type plants helps to refix CO₂ released from BS cells more efficiently by shortening the diffusion pathway between M and BS cells. The quantitative equilibrium of tissues between M and BS cells reflects the biochemical function of C₄ subtypes. The decarboxylation site for CO₂ differs among C₄ subtypes, with NADP-ME species donating CO₂ from C₄ acids in BS chloroplasts and NAD-ME species in BS mitochondria. C₄ NADP-ME species have a higher photosystems I/II ratio and accumulation of NADH dehydrogenase-like complexes in BS. *Cytb6f* was more abundant in C₄ NAD-ME species, which develop slowly compared to NADP-ME crops but are more drought-tolerant and water-efficient (Ghannoum et al. 2002; Vivitha and Vijayalakshmi 2015).

Photosynthetic activity The net assimilation rate (NAR) measures the efficiency of a plant's assimilatory apparatus. It is not a measure of respiration but rather an increase in total dry weight per unit leaf area. Water stress has been

found to decrease plant growth rate (PGR) and mean leaf area (MLA) in common millet, little millet, and foxtail millet (Matsuura et al. 2012). Among the minor millets, barnyard millet recorded the highest NAR value (1.52 mg g⁻¹ day⁻¹), followed by finger millet (1.24 mg g⁻¹ day⁻¹) and a little millet (1.19 mg g⁻¹ day⁻¹) when drought was imposed at the reproductive stage (Senthil et al. 2018). NAR is a gravimetric measurement of the photosynthetic rate as measured by an infrared gas analyzer, with leaf area making a larger contribution (69.3%) to NAR than grain yield (30.7%) in millets (Nanja Reddy et al. 2019; Nanja Reddy 2020).

Under drought stress, gas exchange, net photosynthetic rate (Pn), transpiration rate (Tr), and stomatal conductance (Gs) in switch grass, finger millet, and old world blue stem decreased as soil gravimetric moisture content decreased. Pn decreased in two foxtail millet varieties (0–561 and Jingu 3) evaluated 14 days post-anthesis under drought stress (Dai et al. 2012). Flag leaves had higher net photosynthetic rates than secondary and tertiary leaves, with rate reductions mainly attributed to stomatal closure (Dai et al. 2012). Drought stress significantly reduced net photosynthetic rate, maximal quantum yield, PSII efficiency, and yield in hybrid millet compared to normal millet (Lu et al. 2015). Similarly, moderate drought stress over 18 days during the reproductive growth stage decreased Pn by 16.6% in finger millet (Nanja Reddy et al. 2020).

Photosynthetic activity in millets is influenced by various factors such as species, genotype, developmental stage, and C₄ photosynthetic subtype (Matsuura et al. 2012; Xu et al. 2006; Dai et al. 2012; Lu et al. 2015; Senthil et al. 2018; Mohanabharathi et al. 2019; Nanja Reddy et al. 2020; Pavithra et al. 2020; Mwangoe et al. 2022; Yuan et al. 2022). For example, Pn under drought stress was highest in foxtail millet (35.55 µmol CO₂ m⁻² s⁻¹), followed by finger millet (31.98 µmol CO₂ m⁻² s⁻¹), barnyard millet (31.42 µmol CO₂ m⁻² s⁻¹), proso millet (22.98 µmol CO₂ m⁻² s⁻¹), little millet (22.64 µmol CO₂ m⁻² s⁻¹) and kodo millet (14.98 µmol CO₂ m⁻² s⁻¹) (Pavithra et al. 2020). Finger millet genotype GPU28 had the highest rates of photosynthetic activity, mesophyll efficiency, stomatal conductance, internal CO₂ concentration, transpiration rate, and carboxylation efficiency (Nanja Reddy et al. 2020). Tolerant genotypes maintained higher photosynthetic rates under stress due to higher stomatal conductance and transpiration rates (Nanja Reddy et al. 2020). Drought stress decreased Pn by 16.6% among the finger millet genotypes and had a positive, although insignificant, relationship with grain yield (Nanja Reddy et al. 2020). However, drought stress did not impair CO₂ fixation in finger millet genotypes (Mwangoe et al. 2022), while broom corn millet cultivars showed reduced photosynthetic capacity (Yuan et al. 2022). Further research is needed to understand the effects of drought stress on electron transport,

photosynthetic enzyme activity, and carboxylation in minor millets.

Increased antioxidant defence and ROS scavenging

Reactive oxygen species (ROS) production ROS, including hydroxyl radicals ($\text{HO}^\bullet$), hydrogen peroxide (H_2O_2), superoxide anion radical ($\text{O}_2^\bullet$), and singlet oxygen ($^1\text{O}_2$), are produced in response to stress, potentially causing oxidative damage to macromolecules like lipids, proteins, and nucleic acids. ROS also act as signal transducers, triggering cellular responses under drought stress. In the absence of stress, plants rapidly scavenge ROS to limit ROS toxicity (Mittler 2002). In response to stress, ROS overproduction can be studied by assessing malondialdehyde, H_2O_2 concentrations, and electrolyte leakage (E_L) (Table 2). ROS can react with polyunsaturated fatty acids (PUFA), leading to the production of MDA, a chemical marker considered indicative of oxidative stress and injury (Morales and Munné-Bosch 2019). Although studies on finger millet and foxtail millet show that malondialdehyde (MDA) production is affected by a variety of factors, drought-induced MDA concentrations are lower in tolerant genotypes, suggesting less drought-related oxidative damage (Bhatt et al. 2011; Bartwal and Arora 2016; Mukami et al. 2019; Mundada et al. 2020; Naik et al. 2020; Lata et al. 2011; Dai et al. 2012; Xu et al. 2019; Nematpour et al. 2019; Panda et al. 2021).

Hydrogen peroxide (H_2O_2), is a relatively long-lived and small oxygen-derived free radical. It is of great interest due to its ability to pass through biological membranes to various cell compartments, facilitating signalling functions (Noctor et al. 2014). H_2O_2 can be measured with minimal loss and has been used to evaluate oxidative damage and stress at various developmental stages in millets, including finger millet (Bhatt et al. 2011; Bartwal and Arora 2016; Satish et al. 2018), foxtail millet (Lata et al. 2011; Dai et al. 2012; Nematpour et al. 2019) and proso millet (Nematpour et al. 2019). These studies show that drought stress increases H_2O_2 concentrations with higher levels and less scavenging activity observed in susceptible varieties compared to tolerant varieties. Electrolyte leakage (E_L) is a simple quantitative measure of membrane damage caused by drought-induced oxidative stress. E_L is common in various species, tissues, and cell types under drought and oxidative stress and is primarily due to the efflux of anions (Cl^- , HPO_4^{2-} , NO_3^- , citrate $^{3-}$, malate $^{2-}$) that balance the efflux of potassium cations (Demidchik et al. 2014). Tolerant genotypes of barnyard millet (Singh et al. 2018), finger millet (Bhatt et al. 2011; Sudan et al. 2015; Satish et al. 2018), and foxtail millet (Lata et al. 2011) experience lower E_L during drought, indicating greater membrane integrity and a higher tolerance to oxidative stress (Demidchik et al. 2014).

ROS scavenging Plants have both enzymatic and metabolite-based mechanisms for scavenging ROS. Non-enzymatic radical scavengers like ascorbic acid (AsA), glutathione (GSH), carotenoids, mannitol, α -tocopherol, flavonoids, phenolic compounds alkaloids and non-protein amino acids are important in metabolite-based ROS scavenging. Enzymatic scavenging involves antioxidant enzymes like catalase (CAT), polyphenol oxidase (PPO), superoxide dismutase (SOD), guaiacol peroxidase (GPX), ascorbate peroxidase (APX), dehydroascorbate reductase (DHAR), monodehydroascorbate reductase (MDHAR), and glutathione reductase (GR). Non-enzymatic antioxidants work in a coordinated fashion with antioxidant enzymes to inhibit the overproduction of ROS. AsA and GSH play a significant role in the ascorbate–glutathione (AsA–GSH) or Halliwell–Asada cycles to scavenge ROS and redox homeostasis (Asada 1994). ROS formation is primarily found in mitochondria and chloroplast electron transport chains. In thylakoids, ROS formation results from the over-reduction of photosystems (PS) and electron surplus from light-harvesting chlorophyll protein complexes (LHCP). Carotenoids, scavenge harmful free radicals and protect LHCPs and thylakoid membrane stability (Hussain et al. 2019). Several reports emphasize the role of carotenoids (especially xanthophylls) in the ROS scavenging in millets (Kholová et al. 2011; Ajithkumar and Panneerselvam 2014; Hussain et al. 2019; Naik et al. 2020). Xanthophylls were shown to be spatially associated with chlorophylls where they can account for up to 50% of total carotenoid content and contribute to sustaining the photochemical functions (Hussain et al. 2019). Under water stress, the activities of most APX, CAT and SOD isozymes were also closely related to the total chlorophyll/carotenoids ratio in pearl millet (Kholová et al. 2011).

Of the antioxidant enzymes, SOD is the primary antioxidant enzyme, reacting with superoxide radicals and H_2O_2 to prevent oxidative damage. APX, a key enzyme in the glutathione ascorbate pathway, regenerates NADP^+ and converts H_2O_2 into water. GR and DHAR aid in this reaction. Although the relationship between drought tolerance and ROS scavenging capacity is complex, drought affects the abundance of antioxidant enzymes and metabolites in minor millets. Foxtail millet (Lata et al. 2011), little millet (Ajithkumar and Panneerselvam 2014) and finger millet (Bhatt et al. 2011; Sudan et al. 2015; Bartwal et al. 2016; Naik et al. 2020; Panda et al. 2021) show increased antioxidant enzyme levels during drought stress. In addition, the APX:SOD ratio, a key measure linked to drought tolerance in finger millet, is higher in drought-tolerant than susceptible varieties. Collectively, these findings highlight the importance of ROS scavenging in drought tolerance in millets (Bhatt et al. 2011). The regulation of the antioxidant enzyme genes APX, DHAR, MDHAR, GR, and SOD

was examined in genotypes GP-1 (low Ca^{2+}) and GP-45 (high Ca^{2+}) of finger millet and foxtail millet, respectively (Avashthi et al. 2018). The low Ca^{2+} genotype had higher antioxidant enzyme activity than the high Ca^{2+} genotype, indicating differential regulation of antioxidant enzyme gene expression in finger and foxtail millets. ROS-induced gene expression in finger millet is affected by metal ions, ATP, and DNA binding, among other biological activities (Satish et al. 2018). In particular, the MDHAR and SOD genes were found to modulate ROS-scavenging activity in finger millet in response to drought stress (Avashthi et al. 2018).

Insights into molecular mechanisms of drought tolerance using -omics approaches

The “era of omics” emerged following the development of technologies that allowed the determination of genome sequences at an ever-increasing speed. Omics encompasses various specializations such as genomics, transcriptomics, proteomics, metabolomics, and phenomics, which aim to understand biological processes in light of genomic information and phenotype. Advances in genetics and omics

have explored the genetic basis of complex traits, including abiotic stress tolerance through modifications in DNA, transcript levels, proteins, and metabolites (Muthamilarasan et al. 2019). Omic studies of major crops are rapidly evolving, and genetic improvement is being applied to new challenges revealed by these studies. Here, we review publicly available omics studies addressing molecular mechanisms of drought stress tolerance in minor millets (Table 3).

Genomics

Genomics examines the structural and functional properties of genomes. Early genomic research focused on genome size and physical and genetic mapping. Next Generation Sequencing (NGS) has opened a new era of genomics with the ability to sequence, assemble, and analyze the genomes of plant species within days. The availability of whole genome sequences (WGS) for many major crops has led to the development of genome-assisted breeding (GAB) for crop improvement, the development of new molecular markers and genetic maps, and has facilitated the identification of genes, alleles, and quantitative loci for traits critical to improvement programs. Similarly, transcriptome analysis

Table 3 Drought stress genes studied via transgenic technology

Source millet	Gene/transcription factor	Ectopic expression in crop/plant	Reference
Foxtail millet	<i>SiLEA14</i>	<i>Arabidopsis</i>	Wang et al. (2014)
	<i>SiARDP</i>	<i>Arabidopsis</i>	Li et al. (2014)
	<i>SiATG8a</i>	<i>Arabidopsis</i>	Li et al. (2015)
	<i>SiNF-YA1 and SiNF-YB8</i>	<i>Tobacco</i>	Feng et al. (2015)
	<i>SiASR1</i>	<i>Tobacco</i>	Feng et al. (2016)
	<i>SiLTP</i>	<i>Tobacco</i>	Pan et al. (2016)
	<i>SiASR4</i>	<i>Arabidopsis</i>	Li et al. (2017)
	<i>SiNAC110</i>	<i>Arabidopsis</i>	Xie et al. (2017)
	<i>SiNAC18</i>	<i>Arabidopsis</i>	Dou et al. (2017)
	<i>SiCDPK24</i>	<i>Arabidopsis</i>	Yu et al. (2018)
	<i>SiWLM2b</i>	<i>Rice</i>	Yang et al. (2019)
	<i>SiMYB56</i>	<i>Rice</i>	Xu et al. (2020)
	<i>SiMADS51</i>	<i>Arabidopsis</i>	Zhao et al. (2021)
	<i>SiNRX1</i>	<i>Arabidopsis</i>	Zhang et al. (2022)
	<i>SiMYB53</i>	<i>Arabidopsis</i>	Liu et al. (2023)
	<i>SiNCED 1</i>	<i>Arabidopsis</i>	Huang et al. (2023)
Finger millet	<i>EcNAC1</i>	<i>Tobacco</i>	Ramegowda et al. (2017)
	<i>mt1D</i>	<i>Tobacco</i>	Hema et al. (2014)
	<i>EcDehydrin7</i>	<i>Tobacco</i>	Singh et al. (2015)
	<i>EcbHLH57</i>	<i>Tobacco</i>	Babitha et al. (2015)
	<i>EcbZIP60</i>	<i>Tobacco</i>	Babitha et al. (2015)
	<i>EcGBF3</i>	<i>Arabidopsis</i>	Ramegowda et al. (2017)
	<i>EcNAC67</i>	<i>Rice</i>	Rahman et al. (2016)
	<i>EcbZIP17</i>	<i>Tobacco</i>	Ramakrishna et al. (2018)
	<i>EcCaM</i>	<i>Arabidopsis</i>	Jamra et al. (2021)

has examined differential gene expression under stress conditions, providing insights into gene regulatory mechanisms important in stress responses (Ajeesh Krishna et al. 2022).

Although cereals like rice, maize, barley, and wheat have published genome sequences with full annotations, genome sequencing in minor millets has lagged behind that of many of the major crop species. Complete genome sequences are available for some minor millets, draft sequences have been published for others (e.g., proso and barnyard millet), while some, like little millet and kodo millet have not yet been sequenced (Varshney et al. 2017). A paucity of genomic data makes tools essential for crop improvement, such as DNA marker analysis and genetic mapping, less available for millet than for other cereals. Of those with published WGS, foxtail millet has the smallest genome at 510 Mb, while pearl millet has the largest at 1.76 Gb. Foxtail millet was the first millet crop to have its whole genome sequenced independently for two different genotypes (cv. Zhang Gu × A2 and cv. Zhang Gu × A10) by the US Department of Energy Joint Genome Institute and the Beijing Genomics Institute, respectively (Bennetzen et al. 2012; Zhang et al. 2012).

Two Indian finger millet varieties have been published WGS, ML-365 (Hittalmani et al. 2017) and PR 202 (Hatakeyama et al. 2018). The first draft genome sequence of finger millet, ML-365, was released in 2017, containing 525,759 scaffolds with an average scaffold length of 2275 bp. The sequencing data was analyzed using Illumina and SOLiD sequencing data. The ML-365 genome predicted 85,243 genes, including 2866 drought-responsive genes, 330 calcium transport and accumulation genes, and 146 C₄ pathway genes, using the de novo gene prediction program Augustus (Hittalmani et al. 2017). A draft genome assembly of finger millet was released that is estimated to be 1.5 Gb in size, of which 1189 Mb or 78.2% was covered (Hatakeyama et al. 2018). The genome comprised 2387 scaffolds and 62,348 genes, with 91% functionally annotated and 96.5% single-copy genes. The genome had a maximum sequence length of 5 Mb and a N50 value of 905.318 Kb. The proso millet genome, estimated to be 923 Mb, was submitted to the National Center for Biotechnology Information (NCBI) in 2018 by the China Agricultural University, Beijing (Zou et al. 2019). The assembly of the genome, using short-read Illumina and PacBio long-read sequencing, consists of 5541 contigs, with an N50 contig of 369 kb. The genome contains 55,930 protein-coding genes, 55,527 pseudo-chromosome genes, and 9643 non-coding RNAs. The assembly revealed 58.2% repetitive elements, with 92.1% being transposable elements (TEs). A comparative study revealed that 74.5% of proso millet gene families were shared with four other Poaceae species (Zou et al. 2019). In contrast, only 4.2% of gene families were specific to proso millet (Zou et al. 2019). A draft genome sequence for the barnyard millet genotype STB08 was generated using the Illumina HiSeq

2000 system, covering 90.7% of the predicted genome size of 1.27 Gb (Guo et al. 2017). The data identified 108,771 protein-coding genes, 758 miRNAs, 2,306 transfer RNAs, 1,890 ribosomal RNAs, and other non-coding RNAs. This genome sequence is not yet annotated. Annotated (foxtail millet and finger millet) and draft genome (proso millet and barnyard millet) sequences can identify genes and markers for improving millet growth under biotic and abiotic stresses. Annotated sequences offer potential for comparative genomic studies across millets and related crops.

Chloroplasts are organelles responsible for photosynthesis and play a crucial role in the biosynthesis of starch, amino acids, and fatty acids. The plant chloroplast genome (cp) is a closed circular DNA with a highly conserved quadripartite structure. It contains two inverted repeat sequences (IR), one large single-copy region (LSC), and one small single-copy region (SSC) (Palmer 1985). In recent years, progress has been made in the study of the cp genome of minor millets including foxtail millet (Wang and Zao 2016), little millet (Sebastin et al. 2018), common millet (Cao et al. 2017), Japanese barnyard millet (Sebastin et al. 2019), Indian barnyard millet (Perumal et al. 2016), finger millet (Feng and Gao 2021). The size of the circular cp genome ranges from 135,156 bp (foxtail millet) to 139,929 bp (broomcorn millet) in length with 38.6% (little millet) to 58.6% (foxtail millet) overall GC content. The minor millet cp genomes encode unique 108 (finger millet) to 152 (Indian barnyard millet) functional genes. The DNA sequence of the cp genome can be used as a super barcode or a resource for research in phylogeography, genetic diversity, and evolution. The cp genome could be valuable information for the breeding programs of these millet crops (Feng and Gao 2021).

Publicly available databases

The Foxtail millet Marker Database (FmMDb; <http://www.nipgr.res.in/foxtail.html>), developed at the National Institute of Plant Genome Research (NIPGR) in New Delhi, India, is the first-ever database for *Setaria*, generating large-scale genomic resources after decoding the whole-genome sequence of *S. italica* (Bonthala et al. 2014). The Foxtail Millet Transcription Factor Database (FmTFDb; <http://59.163.192.91/FmTFDb/index.html>) is a comprehensive database that links basic and applied sciences. It contains 2,297 putative transcription factors from 55 families (Bonthala et al. 2014). This database covers genomic location, sequence features, phylogeny, gene ontology (GO), and tissue-specific expression. The Foxtail millet MicroRNA Database (FmMiRNADb; <http://59.163.192.91/FmMiRNADb/index.html>) provides marker information on 355 mature miRNAs, their secondary structure, and potential targets (Khan et al. 2014).

The *S. italica* Functional Genomics Database (SIFGD1) predicts gene function, motif structure and function, regulatory modules, and gene family composition (You et al. 2015). It includes genome, transcript, and protein sequence data from the Beijing Genomics Institute, NCBI, and Phytosome. The Foxtail millet Transposable Element-based Marker Database (FmTEMDb; <http://59.163.192.83/ltrdb/index.html>) was created by Yadav et al. (2015) from 30,706 transposable elements and 20,278 TE-based markers. The complete genome sequence of the foxtail millet is also available in Phytosome (https://phytozomenext.jgi.doe.gov/info/Sitalica_v2_2). The complete and annotated genome sequence of finger millet is now available in Phytosome (https://phytozome-next.jgi.doe.gov/info/Ecora_cana_v1_1). Other public data sources, such as Ensembl plants (<https://plants.ensembl.org/index.html>), Gramene (<https://www.gramene.org/>), PlantsDB (<https://bio.tools/plantsdb>), PLAZA (<http://bioinformatics.psb.ugent.be/plaza/>), GreenPhylDB (<http://www.greenphy1.org/>), SALAD (<http://salad.dna.affrc.go.jp/>), Plant Tribes (<http://fgp.bio.psu.edu/tribedb/>), and PlantgenIE.org (<http://plantgenie.org/>) also contain helpful information for genome mining in millets (Gupta et al. 2017).

Transcriptomics Transcriptomics aims to identify genes involved in biological processes by analyzing and quantifying genome-wide changes in gene expression induced by the environment, stress, or different genetic backgrounds. Previously, methods like northern hybridization, serial analysis of gene expression (SAGE) analyses and microarrays were used for this purpose; however, advancements in analytical tools and next-generation sequencing have transformed the field, offering an alternative to cDNA sequencing. RNA-Seq is a transcriptome analysis technique used to generate genome-wide expression profiles and understand gene regulation under various stress conditions (Wolf 2013; Muthamilarasan et al. 2019).

Transcriptomic analyses of water-stressed plants have identified numerous candidate genes, both regulatory and functional. Genes coding for regulatory proteins include transcription factors (TFs), protein kinases, and phosphatases, which regulate transcription and signal transduction. Genes coding for proteins that protect plant cells from stresses have been annotated as being involved in the biosynthesis of osmoprotectants and antioxidants but also include genes coding for dehydrins, senescence-related genes, heat shock proteins, membrane protectors and transporter proteins (Lata et al. 2011). Transcriptomic analysis has identified numerous drought-induced multi-function genes in foxtail millet (Lata et al. 2010; Pan et al. 2016; Tang et al. 2017; Shi et al. 2018; Xu et al. 2019), proso millet (Yue et al. 2016; Hou et al. 2017; Wang et al. 2017; Zhang et al. 2019; Das et al. 2020; Kotrade et al. 2020) and finger millet

(Hittalmani et al. 2017; Parvathi and Nataraja 2017; Parvathi et al. 2019). Transcriptome research in barnyard millet (Jayakodi et al. 2019) and kodo millet (Suresh et al. 2022) has been more limited.

miRNA MicroRNAs (miRNAs) are small, non-coding, single-stranded RNAs ranging from 21 to 24 nucleotides, primarily involved in post-transcriptional gene regulation. Their highly conserved nature has made miRNAs a significant focus in growth, development, and regulatory stress studies (Sunkar et al. 2017). miRNAs play a crucial role in regulating drought tolerance in plants. In silico approaches have been used to identify novel miRNAs and gene targets in pearl millet, sorghum, foxtail millet, finger millet, and proso millet (Chakraborty et al. 2020). The study identified 1002 miRNAs from 280 families, regulating 23,158 targets across five millet species. 84 miRNAs were conserved in other species, with 215 miRNAs controlling 155 drought-responsive genes, including transcription factors and protein families, revealing the genetics of drought tolerance in millets. These genes could potentially be used to develop functional markers for drought-resistant crop improvement (Chakraborty et al. 2020).

Transcription factors

Transcription factors (TFs) play a crucial role in plant abiotic stress responses by regulating the expression of target genes (Jeyasri et al. 2021). Of the 85 known TF families, only a few have been comprehensively studied for their role in stress tolerance (Baillo et al. 2019; Jeyasri et al. 2021). In millets, TFs are crucial in determining the sensitivity of genotypes to abiotic stress (Abdel-Ghany et al. 2020; Pan et al. 2020). Drought tolerant genotypes in foxtail millet differentially express TFs, enabling precise metabolic tuning under stressful conditions (Puranik et al. 2011; Yu et al. 2020). A study by Qin et al. (2020) found pre- and post-drought differential expression of 298 and 170 genes in millet roots and leaves, respectively, with research focusing on WRKY, NAM/ATAF1/2/CUC1/2 (NAC), dehydration responsive element binding protein (DREB), APETALA2/ethylene responsive factor (AP2/ERF), basic/helix-loop-helix (bHLH), abscisic acid stress ripening (ASR), v-myb avian myeloblastosis viral oncogene homolog (MYB), and basic leucine zipper (bZIP) transcription factors. WRKY TFs exert various physiological functions in plants, including responses to abiotic stresses (Li et al. 2020). In *S. italica*, about 105 *SiWRKY* genes and gene products have been identified (Zhang et al. 2017).

The DREB TF family in foxtail millet contains 65 genes, having expanded through segmental and tandem duplication (Shi et al. 2018). *SiDREB2* is crucial in salinity and dehydration responses (Lata et al. 2011), and an allele-specific

SiDREB2 marker (ASM) has been developed and validated for dehydration stress tolerance (RWC) in foxtail millet (Lata and Prasad 2014). *SiARDP*, an ABA-responsive DRE-binding protein, is differentially expressed under drought conditions in foxtail millet (Li et al. 2014). Similarly, *SiDREB2L* and *SiAP2/ERF* TFs are differentially expressed in millets to a variety of insults, including drought, cold, and salt stress (Lata et al. 2014; Pandurangaiah et al. 2016). NACs regulate stress responses through ABA-dependent and ABA-independent pathways (Puranik et al. 2011; Dou et al. 2017). *SiNAC* genes are differentially regulated in response to drought, salinity, or cold, with some *SiNAC* gene products differing from the typical NAC protein structure (Puranik et al. 2011). In foxtail millet, protein *SiNAC18* regulates gene expression in oxidative stress responses. (Dou et al. 2017).

Proteins of the bHLH TF family contain conserved helix–loop–helix domains and are differentially expressed in response to drought. A study by Wang et al. (2018a, b) identified 149 bHLH genes in foxtail millet and found 8 *SibHLH* differentially expressed under drought stress. Absciscic acid stress ripening (ASR) proteins are plant-specific TF that regulates abiotic stress responses during drought and fruit ripening (Konrad and Bar-Zvi 2008). *SiASR1* in foxtail millet enhances resistance to drought and oxidative stress by regulating ROS homeostasis (Feng et al. 2015). Basic leucine zipper (bZIP) proteins are a prominent TF family in plants. Only one bZIP has been characterized in millets, i.e., *EcbZIP17* from finger millet, which is heat-inducible and increases heat tolerance in transgenic tobacco plants (Ramakrishna et al. 2018). MYB genes are drought-responsive TFs reported in finger millet (Jadhav et al. 2018). Foxtail millet contains 209 *SiMYB* genes (Muthamilarasan et al. 2014a), eleven showing enhanced transcription under salt and drought stress.

The foxtail millet genome includes 124, 27, 39, and 47 genes encoding C_2H_2 type of zinc finger TFs (Muthamilarasan et al. 2014b), trihelix transcription factors (TTF) (Wang et al. 2018a, b), nuclear factor Y (NF-Y) (Feng et al. 2015), and homeodomain leucine zipper (HD-Zip) TFs, respectively. *SiHDZ29* and *SiHDZ45* expression is induced by salinity, drought, and abscisic acid (ABA) (Chai et al. 2018). Nine *SiC₂H₂* genes show differential expression under salinity, dehydration, and cold stress. *SiNF-YA1* and *SiNF-YB8* are induced by ABA, H_2O_2 , and abiotic stress and are TFs controlling the expression of drought-responsive and antioxidant genes, thereby enhancing abiotic stress tolerance (Bishnoi et al. 2023).

Genome sequencing and bioinformatics have identified putative millet-specific TFs, leading to the development of the foxtail millet transcription factor database (FmTFDb; Bonthala et al. 2014). Future TF-specific databases are expected for other millets, including finger millet. TFs

WRKY, NAC, DNA binding with one finger (DOF), DREB, MYB, myelocytomatosis (MYC), bHLH, ASR, and bZIP play important regulatory roles in millet gene expression, controlling downstream genes in response to a variety of environmental stresses, including drought. Identifying and characterising these and other new TFs will be an important but significant challenge to developing more drought-resistant millets (Bishnoi et al. 2023).

Genes

Transcriptome studies have demonstrated differential gene expression in response to water stress, with suppression subtractive hybridization (SSH) analysis identifying dehydration-responsive transcripts in foxtail millet cultivars (Lata et al. 2010; Pan et al. 2016). In addition, a lipid transfer protein, *SiLTP*, was reportedly involved in drought and salt responses in foxtail millet. A study of drought-tolerant *S. italica* cultivars *Yugul* and *An04* revealed altered gene expression in phytohormone metabolism, signalling, detoxification, transcription factors, and stress-related proteins (Tang et al. 2017). RNA-seq analysis of drought-tolerant *S. italica* cultivar M79 and its parent lines revealed that differentially expressed genes (DEGs) form a regulatory network involving various biological processes, including photosynthesis, signal transduction, transcriptional regulation, redox regulation, hormonal signalling, and osmotic regulation. Some photosynthesis-related DEGs were highly expressed in M79 under drought stress compared to parental lines (Shi et al. 2018).

RNA-seq analysis of *S. italica* under PEG-6000 induced water-deficit identified gene expression patterns tied to ABA-response, proline and soluble sugar synthesis, ROS metabolism, channel protein functions, and induction of transcription factors. Four genes of the phenylpropanoid-related pathway were identified in germinating foxtail millet in response to PEG-induced osmotic stress (Xu et al. 2019). Finger millet cultivar GPU-28 exhibited clearly identifiable patterns of drought-induced gene expression, with activating transcription factor6 (ATFP6), phosphatase 2A, RISBZ4, and farnesyl pyrophosphate synthase being key genes in the response (Parvathi et al. 2013). Transcriptome profiling studies have validated the millet drought-responsive gene, *EcDehydrin7*, using qRT-PCR (Singh et al. 2014).

TATA-binding protein (TBP)-Associated Factor6 (*EcTAF6*), a drought response regulatory gene in finger millet, was identified through cDNA library screening (Parvathi and Nataraja 2017). Transcriptome data generated from leaf tissue of GPU-28 finger millet revealed a differential abundance of 1790 transcripts between control and drought-stress treatments (Parvathi et al. 2019). The analysis showed activation of various drought-stress signalling cascade genes, including serine–threonine protein phosphatase 2A (PP2A),

calcineurin B-like interacting protein kinase31 (CIPK31), farnesyl pyrophosphate synthase (FPS), and signal recognition particle receptor α (SRPR α). Housekeeping regulatory genes like TBP-associated factors (TAFs) were also drought-responsive, suggesting that genes involved in basal regulatory processes are activated under drought stress in finger millet (Parvathi et al. 2019).

The leaf transcriptomes were used to study drought tolerance mechanisms in proso millet (Zhang et al. 2019). A cDNA library from drought-tolerant Neimi 5 (NM5) and drought-sensitive Jinshu 6 (JS6) cultivars was sequenced using the Hiseq 4000 sequencing platform. 1695 DEGs were reported when comparing JS6 to NM5 without PEG-induced water stress, and 833 and 2166 DEGs when the transcriptomes of plants stressed using 20% PEG-6000 were compared to control plants. Compared to JS6, different regulation strategies were observed in the jasmonic acid signal transduction pathway of NM5.

The transcriptomes of three *Panicum* species (*P. bisulcatum*, *P. laetum*, and *P. turgidum*) were studied during exposure and recovery from severe drought (Kotrade et al. 2020). The drought-induced pathways included abscisic acid synthesis, signal transduction, osmolyte synthesis, macromolecule protection, cell wall and cuticle remodelling genes, and drought-induced dormancy. In addition, 150 genes were upregulated in the more drought-tolerant *P. laetum* and *P. turgidum*, possibly contributing to their higher drought tolerance (Kotrade et al. 2020).

RNA-seq analysis of kodo millet cultivar 'CO3' identified dehydration-responsive genes in comparing stress and control transcriptomes (Suresh et al. 2022). The study identified 9201, 9814, and 2346 DEGs in 0 vs. 3 h, 0 vs. 6 h, and 3 vs. 6 h libraries, respectively. DEGs were associated with known stress-tolerance pathways, including hormone metabolism, signalling, antioxidant scavenging, photosynthesis, and cellular metabolism. Further studies are needed to characterize these genes and to understand their roles in the dehydration stress responses (Suresh et al. 2022). The transcriptomes of Indian barnyard millet (*E. frumentacea*) compared with its wild relative Japanese millet (*E. crus-galli*), revealed 4159 protein-coding and 2258 RNA transcripts differentially expressed between the two. 3489 protein-coding transcripts were reported unique to Indian barnyard millet (Jayakodi et al. 2019). The study concluded that photosynthesis-induced gene expression is important in drought adaptation in Indian barnyard millet.

Functional validation of genes/transcription factors

Recent transcriptome studies have identified key genes responsible for drought tolerance in millets. Transgenic and genome editing technologies like clustered regularly interspaced short palindromic repeats (CRISPR)/Cas9 have been

developed for crop improvement and functional validation, but prior knowledge of gene function is required (Maharajan et al. 2022).

Transgenic technology Transgenic methods are a popular reverse genetics approach for functional validation of candidate genes. Transgenic methods have functionally validated abiotic stress-responsive genes and TF genes in foxtail millet, finger millet, *Arabidopsis*, rice, and tobacco. Transgenic technology has been instrumental in understanding drought tolerance in plants, including minor millets (Table 4). The *SiLEA14* identified in foxtail millet, which, compared to wild type, induced higher salt and osmotic stress tolerance and improved growth under normal and stress conditions in transgenic *Arabidopsis* seedlings (Wang et al. 2014). A DREB-binding protein gene (*SiARDP*) and *SiATG8a*, an autophagy-related gene were cloned from foxtail millet (Li et al. 2014, 2015). Constitutive expression of *SiARDP* and *SiATG8a* improved drought and salt tolerance in *A. thaliana*, resulting in higher survival rates, chlorophyll concentrations, and proline content. The activation of the nuclear factor Y genes *SiNF-YA1* and *SiNF-YB8* was reported in foxtail millet leaves and roots under drought and salt stress (Feng et al. 2015). Overexpression of *SiNF-YA1* improved drought and salt tolerance in tobacco by maintaining RWC, increasing chlorophyll concentrations, and inducing expression of the antioxidant enzymes superoxide dismutase (SOD), peroxidase (POD), and catalase (CAT). Similarly, *SiNF-YB8* expression led to drought and osmotic tolerance in transgenic lines under stress.

Lipid transfer proteins (LTPs) are cysteine-rich soluble proteins involved in flower, seed, and cuticular wax development. A non-specific LTP gene (*SiLTP*) isolated from foxtail millet and ectopically expressed in tobacco led to higher salt and drought tolerance (Pan et al. 2016). Abscissic acid stress ripening proteins (*SiASR*), a class of plant-specific transcription factors, play crucial roles in plant development, growth, and abiotic stress responses. Overexpression of *SiASR1* in transgenic tobacco significantly increased drought and oxidative tolerance as demonstrated by developmental and physiological analyses of germination rate, root growth, survival rate, relative water content, ion leakage, chlorophyll content, and antioxidant enzyme activities (Feng et al. 2015). A study found that *SiASR4* transcripts increased in foxtail millet seedlings after ABA, NaCl, and PEG treatment. *SiASR4*-transgenic *Arabidopsis* and *SiASR4*-overexpressing foxtail millet showed enhanced drought and salt stress tolerance. The transcription of stress-responsive and ROS scavenging-associated genes was activated in *SiASR4* transgenic plants (Li et al. 2016). The function of *SiNAC18* was analyzed by overexpression in *A. thaliana* (Dou et al. 2017). Significant differences in germination rates between transgenic and wild-type seeds

Table 4 Summary of proteomic studies to dissect drought tolerance in minor millets

Millet crop	Genotypes/tissue	Experimental procedures	Results	Reference
Tef	Leaves	iTRAQ labelling, protein extraction, trypsin digestion, ESI-Q-TOF-MS/MS analysis	A total of 211 differentially regulated proteins identified Putative functions: signaling, transport, cellular homeostasis, pentose metabolism, ROS-producing activities, manganese transport, sugar synthesis, cell wall catabolism	Kamies et al. (2017)
Foxtail millet	Yugul drought resistant seedlings	Protein extraction, trypsin digestion and LC-MS/MS analysis	A total of 321 of differentially regulated proteins identified Putative functions: stress and defense responses, photosynthesis, carbon metabolism, ROS scavenging, protein synthesis, fatty acid and amino acid metabolism, polyamine biosynthesis, hormone metabolism, cell wall modification	Pan et al. (2018)
	Longgu6 Hunagjimmiao/Seeds	2DE, MALDI-TOF/TOF	A total of 83 differentially regulated proteins identified Putative functions: storage, amino acid and protein metabolism, defense and stress responses	Xu et al. (2020)
	Hunagjimmiao/ Seeds	2DE, MALDI-TOF/TOF	A total of 104 differentially regulated proteins identified Putative functions: storage, protein folding, carbohydrate metabolism, glycolysis/gluconeogenesis, amino acid synthesis, detoxification, defense, protein degradation, TCA cycle metabolism, protein synthesis, energy metabolism, transport, pentose phosphate metabolism, signal transduction	Li et al. (2019)
Finger millet	Leaves	TMT/iTRAQ labelling, protein extraction, trypsin digestion, UPLC-MS/MS analysis	A total of 3009 differentially abundant proteins were identified Putative functions: hydrolase activity, glycosyl bond formation, oxidoreductase activity, carbohydrate binding, unsaturated fatty acid and biosynthesis	Li et al. (2021a, b)

suggest that *SiNAC18* positively regulates plant germination under drought conditions through ABA and oxidative stress response signalling pathways in foxtail millet. In foxtail millet, calcium-dependent protein kinases (CDPKs) are crucial for resistance to abiotic stress, with *SiCDPK24* having the highest expression levels under drought stress. Overexpression of *SiCDPK24* enhances drought resistance and survival rate in *Arabidopsis* by activating stress-related genes, such as *RD29A*, *RD29B*, *RD22*, *KIN1*, *COR15*, *COR47*, *LEA14*, *CBF3/DREB1A*, and *DREB2A* (Yu et al. 2018).

The LIM (Lin-11, Isl-1, and Mec-3 domain) gene is a transcriptional factor widely distributed in plants. LIM proteins are crucial for various life activities, including gene expression, cytoskeleton construction, signal transduction, and metabolism. Overexpression of the foxtail millet gene *SiWLIM2b* in transgenic rice resulted in higher survival rates, water content, and less cell damage under drought conditions (Yang et al. 2019). A drought-induced transcription factor, *SiMYB56* discovered in foxtail millet (Xu et al. 2019). Overexpression of *SiMYB56* enhances drought tolerance in transgenic rice, with lower MDA content and higher lignin content under drought conditions. Seventy-two MADS-box transcription factors (*SiMADS*) were identified in the foxtail millet genome, revealing their phylogeny, chromosomal location, and potential functions (Zhao et al. 2021). Transgenic *Arabidopsis* and rice plants overexpressing *SiMADS51* showed reduced drought stress tolerance, lower survival rates, and poorer growth performance. *SiMADS51* is a negative regulator of drought stress tolerance in plants, down-regulating the expression of stress-related genes and affecting plant growth (Zhao et al. 2021).

Thioredoxins (TRXs) are proteins with redox activity that significantly contribute to plant growth, development, and stress resistance. Thirty-five *SiTRX* genes were identified in foxtail millet, with overexpression of *SiNRX1* improving *Arabidopsis*' drought and salt stress tolerance, leading to higher survival rates and improved growth (Zhang et al. 2022). Overexpressing lines generally had higher expression levels of stress-related genes than the wild type. The role of 9-cis-epoxycarotenoid dioxygenase, *SiNCED1* studied in the drought response of foxtail millet (Huang et al. 2023). Overexpression of *SiNCED1* was reported to improve drought resistance by increasing ABA levels and promoting stomatal closure. The overexpression of *SiMYB53* in *A. thaliana* enhanced drought tolerance, insensitivity to ABA, and earlier flowering (Liu et al. 2023). NAC proteins (NAM, ATAF1-2, and CUC2) are plant-specific transcription factors involved in various processes, including growth, development, and stress tolerance. *EcNAC1* was induced under water deficit and salt stress in finger millet (Ramegowda et al. 2017). Transgenic tobacco expressing *EcNAC1* had increased root growth and recovered quicker from various abiotic stresses. Similarly, overexpression of *Ecdehydrin7*,

a drought-responsive gene from finger millet, was directly linked to improved drought tolerance in transgenic tobacco (Singh et al. 2015). Finally, transgenic rice plants expressing *EcNAC67* had higher root and shoot biomass, higher RWC, higher grain yields, better revivability, and lower drought-induced spikelet sterility than non-transgenic plants (Rahman et al. 2016).

Basic leucine zipper (bZIPs) transcription factors regulate plant growth and development. A stress-responsive bZIP, *EcbZIP60*, was cloned from finger millet and functionally validated in tobacco, showing improved tolerance to drought stress, higher stomatal conductance, higher photosynthetic rates, and enhanced growth (Babitha et al. 2015). *EcbZIP60* expression also correlated with significant upregulation of unfolded protein-responsive pathway genes like *BiP1*, *CRT1*, and *PDIL* under normal and stressful conditions (Babitha et al. 2015). A novel bZIP transcription factor (*EcbZIP17*) is characterized in finger millet (Ramakrishna et al. 2018). Transgenic tobacco overexpressing *EcbZIP17* upregulated brassinosteroid signalling genes and showed better growth and grain yield than wild-type (WT) plants under optimal conditions. Under stress, *EcbZIP17*-transformed plants had higher germination rates, higher biomass, better root development, and quicker recovery rates. When exposed to endoplasmic reticulum (ER) stress inducers, transgenic *EcbZIP17* expression resulted in increased leaf diameter, increased height, and higher expression of ER stress-responsive genes *BiP*, *PDIL*, and *CRT1* than WT plants (Ramakrishna et al. 2018).

The transgenic finger millet expressing the bacterial mannitol biosynthetic pathway gene mannitol-1-phosphate dehydrogenase (*mtlD*) exhibited better growth under drought and salinity stress than the wild type (Hema et al. 2014). These plants also had better osmotic adjustment and chlorophyll retention under drought stress. Overexpression of transgenic drought-responsive candidate genes has provided valuable functional insights at the molecular, biochemical, physiological, and phenotypic levels compared to wild-type controls. Numerous stress-responsive genes in millets are viable candidates for transgenic analysis. Functional validation of these genes and evaluation of transgenics in the field will be important aspects of future research.

Genome editing (GE) Targeted genome editing has revolutionized crop improvement, allowing for the editing of desired DNA bases in gene functional studies. The use of site-specific nucleases and CRISPR/Cas9 technologies to alter genes of interest has been used to study traits of agronomic importance in cereals and other crops, including minor millets (Ceasar 2021). However, to date, only a few genes have been functionally characterized in millets using editing approaches. Lin et al. (2018) used CRISPR/Cas9 to mutate the phytoene desaturase gene in foxtail millet,

observing the albino phenotype and considering its applications. CRISPR/Cas9 also alters the Less Shattering1 gene (*Les1*) in green foxtail, reducing head shattering. These studies portend gene editing in functional studies of abiotic stress responses in minor millets (Mamidi et al. 2020).

Proteomics

Transcriptomic studies have identified genes important in abiotic stress tolerance. Nevertheless, proteomic research is crucial to understanding gene product function as it is known that gene expression changes are not always correlated with transcript or protein abundance and activity. Proteomics aims to characterize translational and posttranslational regulation mechanisms and complements transcriptome studies in enhancing our understanding of plant responses to abiotic stress (Hu et al. 2015). Advances in protein extraction and isolation, high throughput analysis, techniques for characterizing post-translational modifications and protein–protein interactions, and bioinformatics have enabled scientists to study plant proteome responses to environmental stress in a complex way. Gel-free protein separation and next-generation proteomics techniques like isobaric tags for relative and absolute quantitation (iTRAQ), multidimensional protein identification technology (MudPIT), isotope-coded affinity tags (ICATs), and targeted mass tags (TMTs) have been widely used to study protein function in abiotic stress responses (Hu et al. 2015; Muthamilarasan et al. 2019).

Proteomic studies have been conducted on millets under abiotic stress, focusing on salinity (Veeranagamallaiah et al. 2008; Pan et al. 2021), yield and lodging resistance (Weng et al. 2021), selenium exposure (Liang et al. 2020), temperature (Singh et al. 2022), methyl jasmonate exposure (Sen et al. 2016), and seed storage protein profiling (Roy et al. 2017). High-throughput comparative proteomics has been used in recent studies on teff (Kamies et al. 2017), pearl millet (Ghatak et al. 2016), sorghum (Goche et al. 2020), foxtail millet (Pan et al. 2018; Li et al. 2019; Xu et al. 2019), and finger millet (Li et al. 2021b). Although proteome research has advanced in minor millets, it still lags behind that of major millets and cereal crops. This section summarizes recent developments in proteomics research of drought stress tolerance in minor millets (Table 4).

Eragrostis tef under controlled and dehydration stress conditions was analyzed using iTRAQ and mass spectrometry. 211 proteins involved in drought tolerance were identified, including those functioning in signalling, transport, cellular homeostasis, and pentose metabolism. Proteins functioning in ROS-producing processes such as photosynthesis, manganese transport, sugar synthesis, and cell wall catabolism and modification were less abundant in drought-exposed teff than in controls (Kamies et al. 2017). Tandem mass tagging (TMT) and liquid chromatography coupled

mass spectrometry (LC–MS/MS) were utilized to identify and quantify differentially abundant proteins (DEPs) in foxtail millet seedlings exposed to drought stress (Pan et al. 2018). Three twenty-one proteins changed in abundance (252 upregulated and 69 downregulated) upon exposure to drought including those involved in drought tolerance stress and defence [(ate embryogenesis abundant (LEA), ASR, ns LTPs, and aquaporin], photosynthesis [ribulose biphosphate carboxylase oxygenase (Rubisco), light harvesting complex II (LHCII), and phosphoenolpyruvate carboxylase (PEPC)], signalling (protein kinases and phosphatases), carbon metabolism [glyceraldehyde phosphate dehydrogenase (GAPDH), phosphoglycerate kinase, and fructokinase], protein synthesis and processing [ribosomal protein, elongation factor, heat shock proteins (HSPs), and protease], metabolism [polyamine metabolism, lignin, and glycine betaine (GB) biosynthesis] and transcription (RNA helicases, PPR, and WD 40) (Pan et al. 2018).

Two recent studies have investigated the proteome changes in foxtail millet grains under drought stress (Li et al. 2019; Xu et al. 2019). A study by Li et al. (2019) analysed the "Huangjinmiao" crop and identified 104 DEPs, with 57 up-regulated and 47 down-regulated proteins. DEPs linked to glycolysis and gluconeogenesis (Fructose-bisphosphate aldolase cytoplasmic isozyme-like), energy metabolism (ATP synthase F1 beta subunit), the tricarboxylic acid cycle (putative aconitate hydratase), starch and sucrose metabolism (1, 4, alpha-glucan branching enzyme 2, glucose-1-phosphate adenylyltransferase), protein folding and degradation (protein disulfide isomerase and trypsin inhibitor), and storage (globulin-1 S allele-like protein) were identified. A study by Xu et al. (2019) analyzed proteomic expression under drought stress in Longgu 6 (tolerant) and Huangjinmiao (sensitive) cultivars, detecting 83 proteins involved in a variety of biochemical and metabolic processes, including stress response, defense, chaperones, storage, and amino acid, protein, and energy metabolism (Xu et al. 2019). The transcriptome and proteome were studied in finger millet under drought stress (Li et al. 2021b). They identified 3009 differentially abundant proteins, all linked to drought stress, including those involved in hydrolase activity, glycosyl bond formation, oxidoreductase activity, carbohydrate binding, and unsaturated fatty acid biosynthesis (Li et al. 2021b).

Metabolomics

Primary (nucleotides, amino acids, carbohydrates, and fatty acids) and secondary (alkaloids, terpenoids, and phenolics) plant metabolites are crucial for growth, biotic interactions, and defence (Li et al. 2018). Metabolomics examines changes in the presence and abundance of low molecular weight metabolites during development or in response to internal or external stimuli. Advances in techniques like gas

Table 5 Summary of drought stress metabolite studies of minor millets (are all of the metabolites listed tied to stress? Only those implicated in drought stress/stress should be included here)

Species	Genotypes	Plant part/tissue	Stress/experimental condition	Method used	Types of metabolites	Reference
Foftail millet	NA	Young versus mature stems and leaves	Metabolic reconstruction	LC-MS/MS	100 metabolites; fatty acid synthesis, lignin, cellulose, nitrogen fixation, amino acid synthesis, carbon fixation, C4 metabolism	De Oliveira Dal' Molin et al. (2016)
	Jigu 42, Henggu 13, Henggu 1, Heimuogu	Seeds	Nutritional properties and aroma characteristics	SDE/GC-MS	62 volatile compounds, nonanal, (z)-2 nonenl, (E, E) 2,4, nonadienal, naphthalene dibutylphthalate	Li et al. (2021a, b)
	NA	Germinating seeds	PEG-induced osmotic stress	HPLC	p-coumaric acid, caffeic acid, ferulic acid and sinapic acid	Yu et al. (2020)
	Yugu 2, An04	Root	Salinity; integrative analysis with transcriptomics	LC-ESI-MS/MS	251 metabolites; Lipids, organic acid, phenolic compounds, amino acids, ion channel and antioxidant system	Pan et al. (2020)
	150 accessions	Leaves	Natural variation	LC-ESI-MS/MS	330 metabolites; flavonoids, phenolamides, hydrocin-namoyl derivatives, nucleotides and lipids	Wei et al. (2021)
Proso millet	A2, Zhnagzagu 3,5,6 & 13	Leaves	Natural variation	LC-ESI-MS/MS	673 metabolites; flavonoids, phenolamides, hydrocin-namoyl derivatives, vitamins	Li et al. (2021a, b)
	Jingu21, Qinzhouhuang, Yugu01, Daobaqu, Niu-maobai	Grains	Diversity and specificity of flavonoids	UPLC-ESI-MS/MS	116 flavonoids	Zhang et al. (2021)
	Yidianhong, Nianidiaoya, Dahulliang, Qishanfeng	Grains	Shade colour of grains (white/gray/black/red)	UHPLC-ESI-Q-TRAP-MS/MS	672 metabolites; tryphospho-metabolites, flavonoids (iso-flavonoid, flavones, flavonol biosynthesis, phenolic acids and flavonoids)	Li et al. (2021a, b)
Little millet	OLM-203/Tarini	Leaves versus roots in seed-lings	Drought	GC-MS	25 metabolic pathways; galactose metabolism, fatty acid metabolism, starch and sucrose metabolism, vitamin K metabolism, pentose phosphate pathway, beta-oxidation, citric acid cycle, fructose and mannose metabolism, alpha linoleic acid and linoleic metabolism, butyrate	Dhawale (2022)

chromatography–mass spectrometry (GC–MS), nuclear magnetic resonance (NMR), flow infusion electrospray ionization high-resolution mass spectrometry (FIE–HRMS), and ultra-high-resolution performance chromatography–mass spectrometry (UHPLC–MS) are accelerating metabolite fingerprinting in plants (Ajeesh Krishna et al. 2022).

Metabolic studies of C_4 metabolism (De Oliveira Dai'Molin et al. 2016), nutritional metabolism, volatile compound biosynthesis (Li et al. 2021a; Yang et al. 2021), natural genetic variation (Wei et al. 2021; Li et al. 2021b), and grain flavonoid diversity (Zhang et al. 2021) have been conducted in foxtail millet. Metabolic profiles in Korean proso millet varieties have also been determined (Kim et al. 2013), while there are no metabolomic reports to date in finger millet, barnyard millet, or kodo millet (Vetriventhan et al. 2020). In tandem with transcriptomic studies, metabolomics has revolutionized the study of the genetic and molecular mechanisms underlying metabolic phenotypes (Table 5). This has been particularly useful in investigating the tolerance of foxtail millet to salinity and atrazine stress (Vetriventhan et al. 2020; Li et al. 2018; Sun et al. 2020; Zhang et al. 2021).

A recent study of foxtail millet cultivars Yugu2 and An04 revealed contrasting transcriptomic and metabolomic profiles in response to salt stress (Pan et al. 2020). Transcriptomics revealed 8887 and 12,249 DEGs in Yugu2 and An04, respectively, with 3149 DEGs shared between varieties. Integrative analysis revealed that phenylpropanoid, flavonoid, lysophospholipid, and lignin biosynthesis were crucial in foxtail millet salinity responses. Yugu2's tolerance was attributed to higher ion channel and antioxidant system efficiencies (Pan et al. 2020). A study of phenylpropanoid metabolite biosynthesis in foxtail millet seeds revealed that cinnamic acid accumulates in germinating seeds under PEG stress while p-coumaric acid, caffeic acid, ferulic acid, and sinapinic acid decreased. Five phenolic compounds affected germination and growth, with 1 mM cinnamic acid inhibiting shoot and root growth, while ferulic acid, caffeic acid, sinapic acid, and p-coumaric acid increased root length and root/sprout growth at lower concentrations. These findings suggest that genes and metabolites are important in regulating drought resistance in foxtail millet seed germination (Yu et al. 2020). The metabolomic profiles in drought-tolerant little millet genotype "OLM-203/Tarini" were studied using GC–MS to identify 60 metabolites from 25 pathways under water stress (Dhawale 2022).

Conclusions and future work

This review covers drought resistance strategies in minor millets, providing insights into the growth, developmental, genetic, molecular, and metabolic processes. While impacts

are evident at all developmental stages, the early stages of germination, seedling, and reproductive growth are most sensitive to drought stress. Drought stress affected reproductive growth more than vegetative growth in adult plants, as millets have greater drought resilience during the vegetative phase. To cope with drought stress, millets use three strategies: escape, avoidance, and tolerance. Escape strategies include short crop cycles, developmental plasticity, and remobilization of assimilates to grain following stress alleviation. Root and leaf morphological modifications, alone or in combination, are part of drought avoidance strategies. Improved root traits enhance water uptake, while superior leaf traits reduce water loss, with root and leaf modifications working in tandem to avoid drought stress.

Morpho-physiological, biochemical and molecular processes are important in minor millet drought tolerance responses, including osmotic adjustment, more efficient stomatal regulation, increased transpiration efficiency, C_4 photosynthesis, and increased ROS scavenging capacity. Research has focused on finger millet, foxtail millet, and proso millet, and additional work is needed in these and other minor millets to develop genotypes that have superior drought resistance. Regarding molecular mechanisms, the genome sequences of foxtail millet, finger millet, proso millet, and broomcorn millet have been used to identify genes and proteins involved in drought response. However, molecular studies have primarily focused on foxtail millet and finger millet, with less attention given to other minor millets. Transcriptomic, proteomic, and metabolomic studies are just beginning, and further research is needed to understand key functional genes and pathways involved in minor millet drought resistance. Only a few available validation studies have transgenically expressed genes and proteins important in drought tolerance.

Genetic engineering and breeding have accelerated the development of drought-tolerant crop cultivars for improved yields and food security. In addition, genome editing has been successfully applied to major cereal crops like rice, wheat, and maize, introducing key traits; however, such approaches have not been applied to minor millets despite their rich nutrient profiles and inherent climate change resilience. To date, gene editing has been performed only in foxtail millet, and current genomic resources and transformation systems may hinder editing in millets. High-resolution editing using CRISPR/Cas9 promises to elucidate specific mechanisms involved in drought tolerance in major cereals (Cesar 2021) and minor millets. This review discusses drought resistance strategies in minor millets, highlighting their potential to enhance productivity under rain-fed conditions, ensuring food and nutritional security in climate change scenarios.

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